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# Surgical Management for Upper Urinary Tract Calculi

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## Keywords

calculi; kidney; lithotripsy; nephrolithotomy; percutaneous; ureteroscopy

## Questions

1. The risk of ureteral perforation is greatest with which of the following intracorporeal lithotripsy technologies?
    a. Electrohydraulic lithotripsy (EHL)
    b. Holmium laser
    c. Pulsed-dye laser
    d. Ultrasonic lithotripsy
    e. Ballistic lithotripsy
2. The risk of retrograde stone propulsion is greatest with which of the following intracorporeal lithotripsy technologies?
    a. EHL
    b. Holmium laser
    c. Pulsed dye laser

d. Ultrasonic lithotripsy
e. Ballistic lithotripsy

3. What are the preferred initial power settings for holmium laser lithotripsy of ureteral stones?
   a. 0.6 J, 6 Hz
   b. 0.6 J, 10 Hz
   c. 1.0 J, 10 Hz
   d. 1.2 J, 10 Hz
   e. 1.0 J, 15 Hz
4. Which intracorporeal lithotripsy technology will most efficiently fragment and evacuate renal calculi?
   a. Ultrasonic lithotripsy
   b. Ballistic lithotripsy
   c. Combination ultrasonic/ballistic lithotripsy
   d. Holmium laser
   e. EHL
5. Which intracorporeal lithotripsy technology has the least risk of ureteral perforation?
   a. Ultrasound
   b. Ballistic
   c. Holmium laser
   d. EHL
   e. Erbium laser
6. Energy sources for shockwave lithotripsy (SWL) include all of the following EXCEPT:
   a. electrohydraulic.
   b. holmium laser.
   c. piezoelectric.
   d. electromagnetic.
   e. microexplosive.
7. What is a major disadvantage of ultrasound imaging for SWL?
   a. Inability to visualize ureteropelvic junction (UPJ) stones
   b. Exposure to ionizing radiation
   c. Inability to visualize radiolucent stones
   d. Expense of ultrasonography systems
   e. Inability to visualize ureteral stones

8. Factors influencing the amount of pain during SWL include all but which of the following?
    a. Power level applied
    b. Stone composition
    c. Type of shockwave generator
    d. Shockwave energy density at the point of skin penetration
    e. Stone location
9. Possible mechanisms producing stone fragmentation during SWL include all of the following EXCEPT:
    a. compression fracture.
    b. spallation.
    c. acoustic cavitation.
    d. dynamic fatigue.
    e. vaporization.
10. What percentage of kidneys experience trauma during SWL?
    a. 0% to 20%
    b. 20% to 40%
    c. 40% to 60%
    d. 60% to 80%
    e. 80% to 100%
11. Risk factors that will enhance the bioeffects of shockwaves include all of the following EXCEPT:
    a. patient age older than 60 years.
    b. pediatric age.
    c. stone burden.
    d. preexisting hypertension.
    e. reduced renal mass.
12. The primary insult to the kidney exposed to shockwaves occurs in which of the following tissues?
    a. Blood vessels
    b. Proximal tubule
    c. Renal papillae
    d. Glomerulus
    e. Renal capsule
13. Which anesthetic technique is associated with the greatest likelihood of a successful SWL treatment outcome?

a. General endotracheal
b. Intravenous sedation
c. Epidural
d. Sedation
e. Topical anesthetic

14. Which of the following is an absolute contraindication to percutaneous nephrolithotomy (PNL)?
    a. Morbid obesity
    b. Uncorrected coagulopathy
    c. Neurogenic bladder
    d. Pelvic kidney
    e. Horseshoe kidney
15. Which treatment maneuver will reduce the likelihood of SWL-induced renal injury?
    a. Begin treatment at a high energy level
    b. Treat at a rate of 120 shocks per minute
    c. Treat with a topical local anesthetic
    d. Pretreat the targeted kidney at a low energy level and then ramp up treatment to a high energy level
    e. Pretreat the contralateral kidney at a high energy level and then ramp up treatment of the target kidney to a high energy level
16. What is the most common secondarily infecting organism after percutaneous stone removal?
    a. *Proteus mirabilis*
    b. *Klebsiella oxytoca*
    c. *Pseudomonas aeruginosa*
    d. *Staphylococcus epidermidis*
    e. *Enterococcus (Streptococcus) faecalis*
17. Which of the following is the antimicrobial of choice for ureteroscopy?
    a. First-generation cephalosporin
    b. Second-generation cephalosporin
    c. Aminoglycoside
    d. Fluoroquinolone
    e. Nitrofurantoin

18. What is the preferred site of puncture into the renal collecting system during access for PNL?
    a. Upper pole infundibulum
    b. Anterior lower pole calyx
    c. Posterior lower pole calyx
    d. Upper pole calyx
    e. Renal pelvis
19. Risk factors for colon injury during PNL include all of the following EXCEPT:
    a. horseshoe kidney.
    b. kyphoscoliosis.
    c. access lateral to the posterior axillary line.
    d. previous jejunoileal bypass for obesity.
    e. upper pole puncture.
20. To minimize the risk of lung and pleura injury during supracostal upper pole access for PNL:
    a. the puncture should be performed during full expiration.
    b. the puncture should be performed during full inspiration.
    c. $CO_2$ should be injected through the ureteral catheter to identify the upper pole calyx.
    d. the puncture should be done with local anesthesia.
    e. the puncture should be performed by a radiologist.
21. Indications for supracostal access during PNL include all of the following EXCEPT:
    a. predominant stone distribution in the upper pole.
    b. access to the UPJ or proximal ureter required.
    c. cystine stones.
    d. multiple lower pole infundibula and calyces containing stone material.
    e. horseshoe kidneys.
22. When performing PNL and endopyelotomy in the same setting, the optimal point of entry is:
    a. posterior upper pole calyx.
    b. posterior lower pole calyx.
    c. anterior upper pole calyx.

d. anterior lower pole calyx.
e. renal pelvis.

23. During access for PNL, what is the preferred initial wire?
    a. Amplatz Super-stiff
    b. Benson
    c. Hydrophilic glide
    d. Lunderquist
    e. J-tipped movable core
24. What is the most common serious error in PNL access?
    a. Not using an Amplatz sheath
    b. Overadvancement of the dilator/sheath
    c. Anterior calyceal puncture
    d. Ultrasonographically guided puncture
    e. The use of telescoping metal dilators
25. What is the appropriate irrigating solution for PNL?
    a. 3% sorbitol
    b. Sterile water
    c. Glycine
    d. Dilute contrast material
    e. 0.9% saline
26. Middle or upper pole access for PNL in horseshoe kidneys is preferred for all of the following reasons EXCEPT:
    a. a higher incidence of retrorenal colon.
    b. malrotation of the renal collecting system.
    c. incomplete ascent of horseshoe kidneys.
    d. anterior medial location of lower pole calyces.
    e. facilitated access to the UPJ or upper ureter.
27. What is the most significant complication of PNL?
    a. Hemorrhage
    b. Extravasation of irrigation fluid
    c. Incomplete stone removal
    d. Urinary tract infection
    e. Pleural effusion
28. What is the risk of arteriovenous fistula formation after PNL?
    a. 1 in 10
    b. 1 in 100

c. 1 in 200
d. 1 in 500
e. 1 in 1000

29. If uncontrolled bleeding persists after nephrostomy tube placement after PNL, what would the preferred approach be?
    a. Insertion of a double-J stent
    b. Administration of furosemide (Lasix) to promote diuresis
    c. Surgical exploration
    d. Immediate angiography
    e. Insertion of a Kaye tamponade balloon
30. If a retroperitoneal injury to the colon is diagnosed after PNL, what is the preferred management?
    a. Surgical exploration and repair
    b. Diverting colostomy with later definitive repair
    c. Leaving the nephrostomy tube in for 2 weeks to allow the tract to mature
    d. Insertion of a double-J stent and withdrawal of the nephrostomy tube into the colon
    e. Immediate removal of the nephrostomy tube
31. The use of double-J stents to reduce the risk of steinstrasse after SWL has been demonstrated to be beneficial for what size of stones?
    a. Greater than 5 mm
    b. Greater than 10 mm
    c. Greater than 15 mm
    d. Greater than 20 mm
    e. Greater than 25 mm
32. Proper management of a stone trapped in a basket, with an avulsed ureter all in continuity and no safety guidewire in place, is:
    a. immediate surgical exploration and primary repair.
    b. cystoscopy to place a guidewire and ureteral stent.
    c. placement of a percutaneous nephrostomy drain.
    d. immediate ureteral reimplantation.
    e. immediate ileal ureter.
33. During the course of a ureteroscopic laser lithotripsy procedure for a 1-cm proximal ureteral stone, a ureteral perforation is noted after

fragmentation and removal of the calculus. On inspection of the perforation, a stone fragment is noted outside the ureter in the retroperitoneum. The most appropriate management is to:

a. terminate the procedure and place a ureteral stent.
b. advance the ureteroscope into the retroperitoneum and remove the stone fragment with a basket device.
c. place a nephrostomy tube.
d. perform laparoscopic exploration and removal of the residual fragment.
e. advance the ureteroscope into the retroperitoneum and fragment the stone with the holmium:YAG laser.

## Answers

1. **a. Electrohydraulic lithotripsy (EHL).** The major disadvantage of EHL is its propensity to damage the ureteral mucosa and its association with ureteral perforation.
2. **e. Ballistic lithotripsy.** Ballistic lithotripsy is accompanied by a relatively high rate of stone propulsion of between 2% and 17% when ureteral stones are treated. **The holmium laser has been associated with a reduced potential for causing retropulsion owing to the weak shockwave that is typically induced during holmium laser lithotripsy.**
3. **a. 0.6 J, 6 Hz.** It is recommended to begin treatment using low pulse energy (i.e., 0.6 J) with a pulse rate of 6 Hz and increase the pulse frequency (in preference to increasing the pulse energy) as needed to speed fragmentation.
4. **c. Combination ultrasonic/ballistic lithotripsy.** Combination ultrasonic and ballistic lithotrites have been reported to provide greater stone clearance rates than do conventional ultrasonic or ballistic lithotrites.
5. **b. Ballistic.** When compared with EHL or ultrasonic or laser lithotripsy, ballistic devices have a significantly lower risk of ureteral perforation.

6. **b. Holmium laser. There are three primary types of shockwave generators: electrohydraulic (spark gap), electromagnetic, and piezoelectric.** Microexplosive generators have also been produced but have not gained mainstream acceptance.
7. **e. Inability to visualize ureteral stones.** Sonographic localization of a kidney stone requires a highly trained operator. Furthermore, localization of stones in the ureter is difficult or impossible.
8. **b. Stone composition.** The discomfort experienced during SWL is related directly to the energy density of the shockwave as it passes through the skin as well as the size of the focal point, parameters that are affected by all of the choices listed except for stone composition.
9. **e. Vaporization. Several potential mechanisms for SWL stone breakage have been described: (1) spall fracture, (2) squeezing, (3) shear stress, (4) superfocusing, (5) acoustic cavitation, and (6) dynamic fatigue.**
10. **e. 80% to 100%.** SWL is now known to induce acute structural changes in the treated kidney in most, if not all, patients. Morphologic studies using both MRI and quantitative radionuclide renography have suggested that 63% to 85% of all SWL patients treated with an unmodified Dornier HM3 lithotripter exhibit one or more forms of renal injury within 24 hours of treatment.
11. **c. Stone burden. Patients with existing hypertension are at increased risk for the development of perinephric hematomas as a consequence of SWL.** Age is a factor on both ends of the scale in that children and the elderly both appear to be at a greater risk for structural and functional changes after exposure to shockwaves. These responses are probably related to a reduction in the large renal reserve present in most healthy adult patients.
12. **a. Blood vessels.** Macroscopically, the acute changes noted in dog and pig kidneys treated with a clinical dose of shockwaves are strikingly similar to those described for patients. This lesion is predictable in size, is focal in location, and is unique in the types of injuries (primarily vascular insult) induced. Regions of damage reveal rupture of nearby thin-walled veins, walls of small arteries, and glomerular and peritubular capillaries, which correlates with

the vasoconstriction measured in both treated and untreated kidneys. These observations show that both the microvasculature and the nephron are susceptible to shockwave damage; however, the primary injury appears to be a vascular insult.

13. **a. General endotracheal.** Patients undergoing SWL with general endotracheal anesthesia experience a significantly greater stone-free outcome than do patients undergoing SWL with alternative anesthetics.
14. **b. Uncorrected coagulopathy.** Uncorrected coagulopathy and an active, untreated urinary tract infection are two absolute contraindications to PNL.
15. **d. Pretreat the targeted kidney at a low energy level and then ramp up treatment to a high energy level.** A number of studies have demonstrated that pretreating the target kidney with low-energy shockwaves, followed by a full clinical treatment dose, will attenuate the renal injury associated with SWL.
16. **d. *Staphylococcus epidermidis* . Cephalosporins are the most appropriately used antibiotics for prophylaxis of surgical procedures in noninfected stone cases, because the most common secondarily infecting organism is *S. epidermidis.***
17. **d. Fluoroquinolone.** The prophylactic antimicrobial agent of choice for ureteroscopy is a fluoroquinolone.
18. **c. Posterior lower pole calyx.** Because the posterior calyces are generally oriented so that the long axis points to the avascular area of the renal cortex, a posterolateral puncture directed at a posterior calyx would be expected to traverse through the avascular zone.
19. **e. Upper pole puncture.** A puncture placed too laterally may injure the colon. The position of the retroperitoneal colon is usually anterior or anterolateral to the lateral renal border. Therefore, risk of colon injury is usually only with a very lateral (lateral to the posterior axillary line) puncture. Posterior colonic displacement is more likely in thin female patients with very little retroperitoneal fat and/or elderly patients, as well as in patients with jejunoileal bypass resulting in an enlarged colon. Other factors increasing the risk of colon injury include anterior calyceal puncture, previous

extensive renal operation, horseshoe kidney, and kyphoscoliosis. A retrorenal colon is more frequently noted on the left side.

20. **a. The puncture should be performed during full expiration.** A supracostal puncture should be performed only during full expiration.
21. **c. Cystine stones.** A supracostal puncture is indicated when the predominant distribution of stone material is in the upper calyces, when there is an associated UPJ stricture requiring endopyelotomy, in cases of multiple lower pole infundibula and calyces containing stone material or an associated ureteral stone, in staghorn calculi with substantial upper pole stone burden, and in horseshoe kidneys.
22. **a. Posterior upper pole calyx.** A posterior upper pole calyx puncture, typically through a supracostal approach, aligns the axis of puncture with the UPJ. This allows the treating urologist to perform endopyelotomy with a rigid nephroscope, while exerting minimal torque on the instrument.
23. **c. Hydrophilic glide.** The hydrophilic glide wire is preferred for entering the collecting system because it is the most flexible and maneuverable wire available.
24. **b. Overadvancement of the dilator/sheath.** Overadvancement of the dilator/sheath is the most common serious error in access for PNL and may result in significant trauma to the renal collecting system and/or excessive hemorrhage.
25. **e. 0.9% saline.** Physiologic solutions should be used for irrigation during PNL to minimize the risk of dilutional hyponatremia in the event of large-volume extravasation.
26. **a. A higher incidence of retrorenal colon.** The optimal point of entry for a horseshoe kidney is through a posterior calyx, which is typically more medial than in the normal kidney because of the altered renal axis and rotation associated with the midline fusion. An upper pole collecting system puncture is often appealing, because the entire kidney is usually subcostal. In most cases the lower pole calyces are anterior and inaccessible percutaneously.
27. **a. Hemorrhage.** Bleeding is the most significant complication of PNL, with transfusion rates varying from less than 1% to 10%.

28. **c. 1 in 200.** Bleeding from an arteriovenous fistula or pseudoaneurysm requiring emergency embolization is seen in less than 0.5% of patients.
29. **e. Insertion of a Kaye tamponade balloon.** If bleeding is not controlled by nephrostomy tube placement and clamping, a Kaye nephrostomy tamponade balloon catheter should be placed (Cook Urological, Spencer, IN). The Kaye nephrostomy tube incorporates a low-pressure 12-mm balloon that may be left inflated for prolonged periods to tamponade bleeding from the nephrostomy tract.
30. **d. Insertion of a double-J stent and withdrawal of the nephrostomy tube into the colon.** Colonic injury is an unusual complication often diagnosed on a postoperative nephrostogram. Typically, the injury is retroperitoneal; thus signs and symptoms of peritonitis are infrequent. If the perforation is extraperitoneal, management may be expectant with placement of a ureteral catheter or double-J stent to decompress the collecting system and by withdrawing the nephrostomy tube from an intrarenal position to an intracolonic position, thus serving as a colostomy tube. The colostomy tube is left in place for a minimum of 7 days and is removed after a nephrostogram or a retrograde pyelogram showing no communication between the colon and the kidney.
31. **d. Greater than 20 mm.** Stents may be particularly advantageous with stones larger than 20 mm.
32. **c. Placement of a percutaneous nephrostomy drain.** Should a ureteral avulsion occur, the patient should undergo immediate diversion of the renal unit with the placement of a percutaneous nephrostomy drain.
33. **a. Terminate the procedure and place a ureteral stent.** When an extruded stone is noted outside the ureter, the procedure should be terminated and a ureteral stent placed.

## Chapter Review

1. EHL produces a hydraulic shockwave and cavitation bubble. It may be used in normal saline solutions.
2. Holmium laser lithotripsy causes stone vaporization by a photothermal mechanism, and when it is used, the stone should be painted.
3. Cyanide may be produced when the holmium laser is used to fragment uric acid calculi. To date, no untoward effects due to this have been reported.
4. Ultrasound breaks the stone by causing the stone to resonate at a high frequency. Considerable heat may develop at the interface.
5. Stone comminution occurs by two basic mechanisms: mechanical stresses produced by the incident shockwave and collapse of cavitation bubbles adjacent to the surface of the stone.
6. The entire ureter can be more easily accessed in the female with a rigid ureteroscope.
7. For uncomplicated ureteroscopies, a ureteral stent may be safely omitted.
8. There is a 3% to 6% incidence of ureteral stricture following ureteroscopy; therefore, follow-up imaging should be performed.
9. Struvite stones must be removed completely to minimize the risk of continued urea-splitting bacteriuria.
10. Cystine and brushite are the stones most resistant to SWL, followed by calcium oxalate monohydrate. Next, in descending order, are hydroxyapatite, struvite, calcium oxalate dihydrate, and uric acid stones.
11. When anticoagulation cannot be temporarily discontinued, the use of ureteroscopy in combination with holmium laser lithotripsy is preferred.
12. The majority of ureteral stones less than 5 mm will pass spontaneously.
13. There are three primary types of shockwave generators: electrohydraulic (Spark Gap), electromagnetic, and piezoelectric. Microexplosive generators have also been produced but have not gained mainstream acceptance.
14. Patients with existing hypertension are at increased risk for the development of perinephric hematomas as a consequence of SWL.

15. Cephalosporins are the most appropriately used antibiotics for prophylaxis of surgical procedures in noninfected stone cases, because the most common secondarily infecting organism is *S. epidermidis*.
16. Transvaginal ultrasonography may be used in the pregnant female to observe the lower ureters.
17. Fifty percent to 80% of pregnant patients will spontaneously pass the calculus.
18. Pregnancy induces a state of absorptive hypercalciuria and mild hyperuricosuria that is offset by increased excretion of urinary inhibitors such as citrate and magnesium, as well as increased urinary output. The metabolic changes in pregnancy do not influence the rate of new stone occurrence. However, paradoxically, it has been suggested that metabolic alterations in urine may contribute to accelerated encrustation of stents during pregnancy.

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# Lower Urinary Tract Calculi

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## Keywords

cystolithotomy; prepucial calculi; prostatic calculi; urethral calculi; vesical calculi

## Questions

1. Which of the following is true about primary bladder calculi?
    a. They are described as those which form without any predisposing cause.
    b. They are more commonly found in well-developed industrialized nations.
    c. Nutritional deficiency has no role to play in the aetiopathogenesis.
    d. They occur in male children in the age group of 10 to 15 years.
2. Which of the following is not true about secondary bladder calculi?
    a. Humans with no infection, obstruction, neurological problems, and absence of foreign body rarely form stones.
    b. Homogenous nucleation is always to be blamed for crystallization in this entity.
    c. Heterogeneous nucleation occurs around the foreign body as a result of infection or due to obstruction with resulting super saturation.

d. Among the neurogenic bladder patients, those with spinal cord injury are more prone to form bladder stones.

3. All of the following are causative factors for stone formation in neurogenic bladder patients except:
    a. completeness of neurological lesion.
    b. urinary tract infection (UTI) by *Proteus, Klebsiella*.
    c. black race.
    d. long-term catheterization.
    e. chronic inflammation.
4. Which among the following statements is false?
    a. Chemodissolution as a sole treatment would be time consuming and not completely efficient.
    b. It is not effective for preventing encrustations over catheters.
    c. This can be considered as the treatment modality as well as a prophylactic measure.
    d. Hemiacridin and Suby solutions causes acidification of urine.
5. Identify the correct statement out of the following?
    a. Extracorporeal shockwave lithotripsy (ESWL) as a modality of treatment has been considered as an option in patients with bladder calculi with artificial urinary sphincters or penile prosthesis.
    b. Transurethral surgery is completely safe even with longer OR times.
    c. Suprapubic surgery is ideal in patients who have had previous urethral or abdominal surgeries.
    d. Stones more than 2 cm should be treated with open surgery.
6. Which of the following statements is false regarding etiology of bladder calculi in patients with urinary diversions?
    a. The most commonly used segment is ileum and colon.
    b. Loss of ileum leads to enteric hyperoxaluria and can lead to formation of stones.
    c. The mucus produced also acts as a nidus for stone formation.
    d. The use of absorbable sutures reduces the incidence of bladder calculi.
    e. Clean intermittent catheterization (CIC) increases the incidence of bladder calculi in comparison with indwelling

catheter.

7. The common metabolic abnormality that results in usage of ileum or colon for diversions is:
    a. hyperchloremic metabolic acidosis.
    b. hypochloremic metabolic alkalosis.
    c. hyponatremic metabolic alkalosis.
    d. hyponatremic metabolic acidosis.
    e. hyponatremic hypokalemic metabolic alkalosis.
8. All of the following are true regarding etiology of bladder calculi in patients with urinary diversions except:
    a. The most common etiological factor is persistence and recurrence of urinary tract infections.
    b. The organisms causing recurrence of infections are *Proteus, Pseudomonas, and Klebsiella*.
    c. The use of ileum causes hematuria dysuria syndrome.
    d. Mucus production leads to formation of biofilm and decreases the penetration of antibiotics.
    e. None of the above.
9. What are the treatment options available for bladder calculi in patients post urinary diversions and augmentation?
    a. Percutaneous approach
    b. Open cystolithotomy
    c. Poucholithotmy
    d. ESWL
    e. All of the above
10. The prevention strategies in case of bladder calculi in patients post urinary diversions and augmentation include all except:
    a. complete clearance of stone.
    b. eradication of infection.
    c. correction of anatomical abnormalities.
    d. use of nonabsorbable sutures.
    e. reduction of mucus formation.
11. Which of the following is false regarding treatment of bladder stone in SCI patients?
    a. ESWL is a treatment option in patients who are prone for autonomic dysreflexia.

b. Laser and transurethral lithotripsy are ideally used for smaller calculi.
c. Long-term use of antibiotics results in reduction of stone formation.
d. Percutaneous or open approach are ideally used for larger calculi.
e. None of the above.

12. The immunosuppressive drug used in transplantation that results in renal calculi is:
    a. antithymocyte globulin (ATG).
    b. mycophenolate Mofetil.
    c. calcineurin inhibitors.
    d. basiliximab.
    e. tacrolimus.
13. Regarding the epidemiology of urethral calculi, which of the following statements is true?
    a. It is more common in females with predilection for the elderly.
    b. It is more common in males with predilection for the elderly.
    c. It is more common in males with predilection for children.
    d. It is more common in females with a bimodal age distribution.
    e. It is more common in males with a bimodal age distribution.
14. Which of the following is not true about migratory urethral calculi?
    a. Migratory urethral calculi are the most common cause of urethral calculi.
    b. Structural or functional obstruction predisposes to urethral calculi.
    c. Concomitant bladder and upper tract calculi are seen in 2% and 18% of patients, respectively.
    d. Concomitant bladder and upper tract calculi are seen in 18% and 2% of patients, respectively.
15. Which of the following is true about migratory urethral calculi?
    a. Migratory urethral calculi are usually solitary and larger than primary calculi.
    b. Anterior urethral stones are more common than posterior urethral stones.
    c. The majority of patients have insidious onset of symptoms.

d. Posterior urethral stones are palpable on digital rectal examination.

16. Which of the following is true about primary urethral calculi?
    a. Primary urethral calculi are so named because they are the primary and common cause of urethral calculi.
    b. Primary urethral calculi form primarily in the urethra irrespective of any obstruction of urethra or urinary stasis.
    c. The presence of a foreign body and urinary infection provide a conducive milieu for stone formation.
    d. The predominant composition of primary urethral calculi is calcium oxalate monohydrate.
17. Which of the following are the most commonly isolated organisms in case of infection stones?
    a. *Proteus* spp.
    b. *Escherichia coli*
    c. Enterococci
    d. All of the above
18. Which of the following is false regarding urethral calculi?
    a. Primary urethral calculi have a protracted course and insidious symptoms.
    b. Painless chronic retention occurs in 78% of patients.
    c. Patients with migratory calculi present immediately in view of sudden and drastic onset of the symptoms.
    d. Fistulous complications are more common in those who cannot report their symptoms due to decreased sensation.
19. The following statements are true about urethral calculi except:
    a. The majority of urethral stones are radiolucent and not visible on plain radiographs.
    b. Retrograde urethrogram can identify the stones as filling defect and has been the mainstay of investigation.
    c. Transrectal ultrasonography can identify the prostatic urethral calculi as a hyperechoic structure with post-acoustic shadowing.
    d. Upper tract stones may coexist and hence advocate an ultrasonography of the kidney, ureter, and bladder.

20. Which of the following is false regarding management of urethral calculi?
    a. Stones located in the posterior urethra can be pushed back into the bladder for extracorporeal shockwave lithotripsy or intracorporeal fragmentation with mechanical or electrohydraulic lithotripsy.
    b. Shockwave lithotripsy after pushback is successful, with a stone-free rate of 60%.
    c. Intracorporeal laser lithotripsy is more traumatic to the mucosa, with a success rate of 70% to 80%.
    d. For stones that do not get pushed back into the bladder, in situ fragmentation using laser or pneumatic lithotripsy should be undertaken.
21. Identify the correct statement out of the following.
    a. For stones in the anterior urethra, pushback into the bladder is hardly ever possible and is not to be attempted.
    b. A small but irregular stone can be milked out of the external meatus.
    c. Holmium laser is more traumatic than lithotripters when used for anterior urethra.
    d. A dorsal meatotomy helps in controlled removal of the stone and actually reduces urethral trauma.
22. Identify the incorrect statement out of the following.
    a. Urethral calculi amount to 0.3% to 1% of all stone disease.
    b. Normal urethra can clear stones up to a size of around 8 to 10 mm.
    c. Despite epilation, hair-bearing grafts lead to hair regrowth in 1% to 2% of patients.
    d. Nearly 22% of urethral stones having some form of lower urinary tract symptom.

# Answers

1. a.**They are described as those which form without any predisposing cause**

Primary bladder stones usually are described as those which form without any predisposing cause, while a secondary cause is found in most patients with certain predisposing causes. The primary bladder stones are historically known to occur associated with nutritional deficiency. These are known to be rampant in undernourished or nutritionally compromised individuals. The cause for formation of these calculi are believed to be a combination of decreased urine output, alteration in the urine pH, and other metabolic abnormalities. Vitamin deficiency and dietary compromise in the form of deficient animal proteins is responsible in the genesis of these stones. The primary bladder stones are known to form within the first 5 years of life and have a male preponderance to them.

2. **b. Homogenous nucleation is always to be blamed for crystallization in this entity.** Humans with no infection, obstruction, neurological problems, and absence of foreign body rarely form stones. Crystal formation has been implicated in formation of these stones. Homogenous nucleation rarely is to be blamed for crystallization in this entity. Heterogeneous nucleation occurs around the foreign body as a result of infection or due to obstruction with resulting supersaturation. Among the neurogenic bladder patients, those with spinal cord injury are more prone to forming bladder stones.
3. **c. Black race**

The factors implicated include complete neurological lesion, urinary tract infection caused by *Klebsiella*, and Caucasian race.

4. **b. It is not effective for preventing encrustations over catheters.**

Chemodissolution as a sole treatment would be time consuming and not completely efficient. In the current era its role is limited to use in select cases as an adjunct treatment. The treatment of chemodissolution is particularly effective for encrustation over long-term catheters. This can be considered as the treatment modality as well as a prophylactic measure.

5. **a. Extracorporeal shockwave lithotripsy (ESWL) as a modality of treatment has been considered as an option in patients with bladder calculi with artificial urinary sphincters or penile prosthesis.**

ESWL as a modality of treatment has been considered as an option in patients with bladder calculi with artificial urinary sphincters or penile prosthesis. Any endoscopic intervention in such a situation is fraught with jeopardizing the integrity of the prosthesis or sphincter device. It has also been considered to be a treatment option in stones in neobladders and medically high-risk patients.

6. **e. Clean intermittent catheterization (CIC) increases the incidence of bladder calculi in comparison with indwelling catheter.**

CIC has been found to reduce the incidence of bladder calculi in comparison with indwelling catheter.

7. **a. Hyperchloremic metabolic acidosis**

The commonly used intestinal segments at present are ileum and colon that are prone to many metabolic abnormalities, which increase the risk of stone formation. The common metabolic abnormality that results in usage of ileum or colon for diversions is hyperchloremic metabolic acidosis, which in turn may cause hypercalciuria due to decrease in absorption of calcium (Assimos, 1996).

8. **c. The use of ileum causes hematuria dysuria syndrome.**

The most common etiological factor is persistence and recurrence of urinary tract infections and the organisms causing recurrence of infections are urease-splitting organisms such as *Proteus, Pseudomonas, and Klebsiella*. Mucus production leads to formation of stones by two mechanisms. It acts as a nidus for stone formation and also helps in

formation of biofilm that decreases the penetration of antibiotics. The use of stomach and not the ileum results in hematuria dysuria syndrome.

9. **e. All of the above.**

All the above-mentioned treatment modalities can be used in treating patients of calculi post bladder diversion and augmentation.

10. **d. Use of nonabsorbable sutures.**

The prevention strategies include mainly complete removal of stone and treatment of infection, correction of anatomical abnormalities, prevention of mucus, and use of absorbable sutures.

11. **c. Long-term use of antibiotics results in reduction of stone formation.**

The use of long-term antibiotics for eradication of infection can lead to reduction of gut bacteria and leads to enteric hyperoxaluria, which in turn leads to stone formation.

12. **c. Calcineurin inhibitors.**

The immunosuppressive drugs causing calculi in post-transplant patients are calcineurin inhibitors and steroids.

13. **e. It is more common in males, with a bimodal age distribution.**

Urethral calculi have been exceedingly more common in males, with a bimodal age distribution. The first peak occurs in early childhood and the second peak incidence occurs in the fourth or fifth decades of life. Shorter length of the female urethra and higher peak flow rates in adolescence and younger age groups may have the protective effect for the younger demographic age group.

14. **d. Concomitant bladder and upper tract calculi are seen in 18% and 2% of patients, respectively.**

A vast majority of urethral calculi are migratory in nature. Any pathology, either structural or functional, affecting the urethra, such as urethral stricture, meatal stenosis, or Benign prostatic hyperplasia, can impede the normal stone passage and result in a migratory calculus, which otherwise would have been cleared. It was commonly believed that the bladder was the source of migratory calculi, but recent evidence has suggested that the source of these migratory calculi could as well be the upper urinary tract calculi. Patients with urethral calculi had concomitant bladder and upper tract calculi in 2% and 18%, respectively.

15. **a. Migratory urethral calculi are usually solitary and larger than primary calculi.**

Migratory urethral calculi are usually solitary and larger than primary calculi. Posterior urethral calculi are more common, as high as 78% in some series. Migratory urethral calculi have a drastic onset of symptoms, with only a very few presenting with lower urinary tract symptoms (LUTS). Digital rectal examination is normal except for minimal tenderness of prostate while anterior urethral calculi may be palpable.

16. **c. The presence of a foreign body and urinary infection provide a conducive milieu for stone formation.**

Primary urethral calculi are named primary as they originate at the same site where they are found ultimately. In reality, they form secondary to obstruction of urethra at any level or urinary stasis in urethral diverticula. Presence of a foreign body and urinary infection provides a conducive milieu for stone formation. Most of these stones are small, round, without a core or nucleus, and the predominant composition is struvite (magnesium ammonium phosphate), although other types such as calcium phosphate and uric acid have been reported.

17. **d. All of the above.**

Most of the primary calculi are small, round, without a core or nucleus, and the predominant composition is struvite (magnesium ammonium phosphate). In the presence of infection stones, the commonly isolated organisms are *E. coli*, *Proteus*, and enterococci.

18. **b. Painless chronic retention occurs in 78% of patients.**

In a large contemporary series of urethral calculi, a vast majority of 78% had acute painful urinary retention and the remaining 22% had some form of lower urinary tract symptoms, reiterating the fact that the majority of urethral stones remain symptomatic (Kamal et al., 2004). Other authors have also observed similar findings, although the rates of urinary retention have varied from 0% to 90% in literature.

19. **a. The majority of urethral stones are radiolucent and not visible on plain radiographs.**

The majority of urethral stones are radio-opaque and visible on plain radiographs. Retrograde urethrogram can identify the stones as filling defect and has been the mainstay of investigation. Transrectal ultrasonography can identify the prostatic urethral calculi as a hyperechoic structure with postacoustic shadowing. A high-frequency (10 MHz) linear transducer placed on the dorsal surface of the penis along its long axis can screen the entire urethra. Upper tract stones may coexist and hence advocate an ultrasonography of the kidney, ureter, and bladder or a cross-sectional computerized tomography, especially if there exists a strong suspicion.

20. **c. Intracorporeal laser lithotripsy is more traumatic to the mucosa, with a success rate of 70% to 80%.**

Stones located in the posterior urethra can be pushed back into the bladder for extracorporeal shockwave lithotripsy or intracorporeal fragmentation with mechanical or electrohydraulic lithotripsy. Shockwave lithotripsy after pushback is successful, with a stone-free rate of roughly 60%. Intracorporeal laser lithotripsy is less traumatic to the mucosa, with a

guaranteed success rate of 85% to 90%. For stones that do not get pushed back into the bladder, in situ fragmentation using laser or pneumatic lithotripsy should be undertaken.

21. **a. For stones in the anterior urethra, pushback into the bladder is hardly ever possible and is not to be attempted.**

For stones in the anterior urethra, pushback into the bladder is hardly ever possible and is not to be attempted. Small stones can be milked out of the external meatus, but irregular and large stones should not be milked as they cause urethral damage. Holmium laser is least traumatic when used for anterior urethra. A ventral meatotomy helps in controlled removal of the stone and actually reduces urethral trauma.

22. **c. Despite epilation, hair-bearing grafts lead to hair regrowth in 1% to 2% of patients.**

Attempts at epilation before engraftment have reduced but not eliminated the risk as the follicles persist and lead to hair growth in 3% to 6% of patients.

## Chapter Review

1. Primary bladder calculi are most common in children younger than the age of 10, with a peak incidence at 2 to 4 years. The disease is much more common in boys than in girls.
2. The most common presentation of bladder stones is hematuria with lower urinary tract symptoms.
3. Children with bladder stones often present with abdominal discomfort, dysuria, frequency, and hematuria.
4. Pulling the penis, in children, is considered pathognomic of bladder stone.
5. The options for treatment of bladder stones are medical management, extracorporeal shockwave lithotripsy, transurethral lithotripsy, suprapubic cystolithotomy, suprapubic cystolithotripsy, and open surgery.

6. The advantages of percutaneous approach in bladder stones are safety and efficacy and potentially lesser risk to the urethra.
7. Urethral calculi amount to 0.3% to 1% of all stone disease, and obstructing urethral calculus is a very rare presentation of lower urinary tract urolithiasis.
8. More commonly urethral calculi patients present with acute painful retention of urine from sudden impaction of the stone.
9. Patients with at least one symptom of prostatitis are 3.2 times more likely to harbor a significant prostatic calculus than age-matched asymptomatic group.
10. The common factors associated with prepucial calculi are severe phimosis along with poor hygiene and low socioeconomic status.
11. Plain x-ray film shows calculi in 14% prepucial calculi in patients.

# PART XI

# Neoplasms of the Upper Urinary Tract

## OUTLINE


96. Benign Renal Tumors
97. Malignant Renal Tumors
98. Urothelial Tumors of the Upper Urinary Tract and Ureter
99. Surgical Management of Upper Urinary Tract Urothelial Tumors
100. Retroperitoneal Tumors
101. Open Surgery of the Kidney
102. Laparoscopic and Robotic Surgery of the Kidney
103. Nonsurgical Focal Therapy for Renal Tumors
104. Treatment of Advanced Renal Cell Carcinoma

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# Benign Renal Tumors

*William P. Parker, and Matthew T. Gettman*

## Keywords

angiomyolipoma; benign renal mass; cystic nephroma; leiomyoma; metanephric adenoma; mixed epithelial and stromal tumor; oncocytoma; papillary adenoma; renal cyst

## Questions

1. A 47-year-old woman is referred for the evaluation of renal cyst disease. What features would suggest a diagnosis of autosomal dominant polycystic kidney disease?
    a. One simple (Bosniak I) renal cyst in each kidney
    b. One simple (Bosniak I) renal cyst in the right kidney and two hepatic cysts
    c. One complex (Bosniak III) renal cyst in the left kidney and one simple (Bosniak I) renal cyst in the right kidney
    d. Two simple (Bosniak I) renal cysts in each kidney
    e. Two simple (Bosniak I) renal cysts in the right kidney
2. The key finding on CT imaging that differentiates Bosniak I-IIF from Bosniak III-IV renal cysts is:
    a. enhancement on administration of intravenous contrast.
    b. high-density cystic fluid.
    c. intraseptal calcifications.
    d. septal nodularity.

e. septal thickening.

3. A 58-year-old man undergoes a partial nephrectomy for a 4-cm renal mass. On pathologic evaluation, the mass is diffusely eosinophilic with invasion of the perirenal fat. Helpful in establishing a diagnosis of oncocytoma using immunohistochemical staining is:
    a. Cytokeratin-7.
    b. Estrogen receptor.
    c. HMB-45.
    d. Melan-A.
    e. WT1.
4. A 25-year-old man presents with a renal mass identified on axial imaging obtained during the evaluation of a pneumothorax. On examination he has multiple raised lesions around his nasal folds. What chromosome is likely altered in this patient given the likely disease?
    a. Chromosome 4
    b. Chromosome 7
    c. Chromosome 9
    d. Chromosome 16
    e. Chromosome 17
5. A 30-year-old woman with neurocognitive delay, epilepsy, and multiple cutaneous lesions presents with her family for the evaluation of bilateral fat containing renal masses. As you discuss options for management, you review the role of everolimus in the management and stabilization of renal mass growth. This medication is targeting what abnormal genetic pathway?
    a. *EHHADH* mutations resulting in decreased adenosine triphosphate (ATP) production
    b. *Hamartin* gene mutations that result in unchecked protein synthesis and cellular growth
    c. Nuclear transcription factor mutation resulting in constitutive activity
    d. *PKD1* gene mutations that impair intracellular calcium regulation
    e. *VHL* gene mutations that result in unregulated angiogenesis

6. A 44-year-old woman is referred for a suspected angiomyolipoma. All of the following radiographic findings are consistent with the diagnosis except:
    a. hyperintensity on T1 MRI sequences.
    b. hyperintensity on T2 MRI sequences.
    c. hyperintense borders on in and opposed phase MRI sequences.
    d. intralesional density of –30 Hounsfield units (HU) on non-contrast CT.
    e. hypointensity of T1 fat-suppressed MRI sequences.
7. The use of everolimus in the management of tuberous sclerosis-related angiomyolipoma is based on phase III data in which a 30% reduction in mass was observed in ____ patients.
    a. 20%
    b. 40%
    c. 60%
    d. 80%
    e. 100%
8. A 42-year-old woman presents with a hypoenhancing renal mass that is resected and determined to be a metanephric adenoma. A paraneoplastic syndrome identified in this disease is:
    a. hyperaldosteronism.
    b. hyperparathyroidism.
    c. Cushing syndrome.
    d. polycythemia.
    e. syndrome of inappropriate antidiuretic hormone secretion (SIADH).
9. A 15-year-old boy is referred to your clinic for the evaluation of a cystic renal mass. On CT, the mass has evidence of multiple septations that enhance on contrast administration. Due to concern for Wilms tumor, a radical nephrectomy is performed. The final pathology is a cystic nephroma. A finding that will aid in differentiation of Wilms tumor from cystic nephroma is:
    a. alpha-methylacyl-CoA racemase (AMACR) positivity.
    b. an absence of blastemal elements.
    c. cysts lined with hobnail epithelium in a background of spindle cell fascicles.

d. no evidence of estrogen receptor positivity.
e. staining for WT-1.

10. A 58-year-old man presents with renal colic and hypertensive urgency. On evaluation he is found to have a cystic mass surrounding the renal capsule and compressing the kidney. This is consistent with a diagnosis of:
    a. cystic nephroma.
    b. leiomyoma.
    c. lymphangioma.
    d. papillary adenoma.
    e. renal hemangioma.
11. A 35-year-old woman presents with hypokalemia and hypertension refractory to medical management. Evaluation reveals a small hypoechoic mass in the right kidney on ultrasound. The etiology of hypokalemia in this patient is related to:
    a. Cushing syndrome.
    b. diuretic use.
    c. hyperaldosteronism.
    d. hyperparathyroidism.
    e. SIADH.
12. A 35-year-old woman in her third trimester is brought to the emergency department unresponsive with hypotension, anemia, and is found to have a retroperitoneal hemorrhage. The most likely cause is:
    a. angiomyolipoma.
    b. clear cell renal cell carcinoma.
    c. metanephric adenoma.
    d. oncocytoma.
    e. trauma.
13. The following are common to most benign renal mass presentations EXCEPT:
    a. female gender.
    b. incidental diagnosis.
    c. older patient age.
    d. smoking history.
    e. stable size.

# Answers

1. **d. Two simple (Bosniak I) renal cysts in each kidney.** According to the diagnostic guidelines for autosomal dominant polycystic kidney disease, the finding of two renal cysts (unilateral or bilateral) in patients less than 30, two cysts in each kidney in patients between 30 and 59(d), and four cysts in each kidney after 60 is suggested of the disease. The other descriptors (a, b, c, e) are not sufficient for the diagnosis.
2. **a. Enhancement on administration of intravenous contrast.** The Bosniak system for classifying renal cysts defines enhancement of the solid cyst components on the administration of intravenous contrast as a concerning finding toward the risk of malignancy and is the key finding that differentiates Bosniak I-IIF from Bosniak III/IV renal cysts (a). The remainder (b, c, d, e) can be seen in Bosniak II and IIF cysts.
3. **a. Cytokeratin-7.** For differentiation of oncocytoma from eosinophilic chromophobe renal cell carcinoma (the diagnostic dilemma in this case), the ISUP (International Society of Urologic Pathology) recommends the use of cytokeratin-7, which is rarely positive in oncocytoma and is diffusely positive in chromophobe renal cell carcinoma (a). Estrogen receptor positivity is seen in multiple other renal tumors (b), HMB-45 (c) and Melan-A (d) are positive in angiomyolipomas, and WT1 is positive in metanephric adenomas (e).
4. **e. Chromosome 17.** The likely disorder is Birt-Hogg-Dubé syndrome, which is associated with mutations in the folliculin gene on chromosome 17 (e). Chromosome 4 (a) is mutated in autosomal dominant polycystic kidney disease; chromosome 7 (b) mutations are associated with papillary adenomas; papillary renal cell carcinoma, and metanephric adenoma; chromosome 9 (c) is mutated in tuberous sclerosis; and chromosome 16 (d) in both autosomal dominant polycystic kidney disease.
5. **b. *Hamartin* gene mutations that result in unchecked protein synthesis and cellular growth.** Hamartin (b) is one of two genes

mutated in tuberous sclerosis, the described disorder. In this disease, mutation results in activation of the mTOR pathway and unregulated growth. The other described mutations relate to Fanconi syndrome (a), a method for castration resistance in prostate cancer (c), polycystic kidney disease (d), and clear cell renal cell carcinoma (e).

6. **c. Hyperintense borders on in and opposed phase MRI sequences. (c).** On these sequences, angiomyolipomas will appear to have a hypointense border ("India ink artifact") due to the presence of intralesional fat.
7. **d. 80%.** The phase III randomized controlled trial of everolimus in the management of TSC-related angiomyolipomas demonstrated a 42% response of 50% or greater reduction in size, 80% response of 30% or greater reduction in size (d), and a 100% rate of stabilization or partial response. Subsequent follow-up to 4 years revealed a 92% any response rate with only 14% experiencing progression.
8. **d. Polycythemia.** Polycythemia (d) has been identified in metanephric adenoma due to tumor production of erythropoietin, IL-6, IL8, G-CSF, and GM-CSF. The remainder (a, b, c, e) of choices describe paraneoplastic syndromes not associated with metanephric adenoma.
9. **b. An absence of blastemal elements.** (b) Blastemal and embryonal elements are common to Wilms tumor and are notably absent in cystic nephroma. While answer choice (c) describes findings common to cystic nephroma, this will not definitively rule out Wilms tumor. AMACR (a) and WT1 (e) staining are used in the diagnosis of metanephric adenomas. Cystic nephromas stain positive for estrogen receptor (d).
10. **c. Lymphangioma.** Of the described choices, lymphangiomas (c) are known to present with cystic extrinsic compression of the kidney due to aberrant lymphatic development in the perirenal lymphatics. Leiomyomas (b), papillary adenomas (d), and renal hemangiomas (e) will not appear cystic and external to the kidney; while cystic nephroma (a) may appear cystic, they are most often within the renal parenchyma.

11. **e. SIADH.** This is a classic example of a juxtaglomerular tumor (JGCT). The source of hypertension in these patients is hyperproduction of renin, which results in a secondary cause of hyperaldosteronism (c). The other choices represent other paraneoplastic processes unrelated to JGCTs.
12. **a. Angiomyolipoma.** The most common cause of Wunderlich syndrome is from an angiomyolipoma (a). An additional clue to this presentation is the patient's pregnancy, where hormonal changes can result in growth of the lesion.
13. **d. Smoking history. All of the answers suggest a benign renal mass presentation, except for smoking (d), which is a risk factor for a malignant renal mass.**

## Chapter Review

1. Renal cyst disease is the most common benign renal tumor.
2. Acquired cystic kidney disease is associated with a significant increase in malignancy risk and should be followed closely.
3. Most cysts require no additional follow-up or therapy.
4. The Bosniak classification is a useful tool to estimate malignancy risk and direct therapy.
5. Renal oncocytoma is the most common benign enhancing renal mass.
6. Oncocytomas are histologically similar to chromophobe renal cell carcinoma (RCC), and immunohistochemical staining may be required to distinguish the two.
7. Oncocytoma is associated with Birt-Hogg-Dubé syndrome: characterized by pulmonary cysts, spontaneous pneumothoraces, and fibrofolliculomas.
8. When suspected, a percutaneous biopsy may provide a diagnosis when core tissue is available for additional immunohistochemical studies.
9. Active surveillance represents a reasonable treatment strategy that minimizes risk and increases certainty based on growth kinetics.
10. Angiomyolipoma is a benign tumor of dysmorphic blood vessels, smooth muscle, and adipose tissue that can be diagnosed definitively

on cross-section imaging due to the presence of macroscopic fat.

11. Most patients present asymptomatically; however, angiomyolipomas (AML) is the most common cause of spontaneous retroperitoneal hemorrhage.
12. Tuberous sclerosis is associated with the development of multifocal and bilateral renal AML and is associated with activation of the mTOR pathway.
13. Treatment depends on size, presence of symptoms, and pregnancy status and should be tailored to the patient with the goal of renal function preservation.
14. The treatment of choice in patients with acute hemorrhage is selective renal angioembolization.
15. Everolimus is indicated for the management of larger, multifocal AMLs in patients with TSC and lymphangioleiomyomatosis (LAM).
16. Papillary adenomas are ≤5mm in size and are therefore not readily diagnosed on imaging.
17. Papillary adenomas share common immunohistochemical staining, genetic changes, and are found in association with papillary RCC, suggesting a commonality between these diseases.
18. Metanephric adenoma is a rare, benign enhancing renal mass. Symptoms are present in approximately 50% of patients, with a female predisposition.
19. Metanephric adenoma may exist on a continuum with Wilms tumors and papillary RCC, and can be distinguished based on histology and immunohistochemical staining.
20. Cystic nephroma and mixed epithelial and stromal tumor (MEST) are biologically related and histologically similar.
21. Frequent expression of estrogen and progesterone receptor in tumor tissue is reported with a female predisposition.
22. If feasible, a nephron-sparing approach with partial nephrectomy is the preferred management strategy.
23. Leiomyomas are described along the urinary tract, but most often from the renal capsule.
24. Radiologic examination is insufficient to confirm the diagnosis.
25. Nephron-sparing approaches are preferred when technically possible.
26. Multiple rare tumors of the kidney have been described.

27. Radiologic differentiation from renal malignancy is not possible.
28. Reninoma, a benign tumor of the renal juxtaglomerular cell apparatus, is an important but rare cause of secondary hypertension and hypokalemia.

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# Malignant Renal Tumors

*Steven C. Campbell, Brian R. Lane,, and Philip M. Pierorazio*

## Keywords

active surveillance; checkpoint inhibitors; chronic kidney disease; partial nephrectomy; radical nephrectomy; renal cell carcinoma; renal mass; thermal ablation; tyrosine kinase inhibitors

## Questions

1. What is the most accurate imaging study for characterizing a renal mass?
    a. Intravenous pyelography
    b. Ultrasonography
    c. Computed tomography (CT) with and without contrast enhancement
    d. CT-positron emission tomography (PET) scan
    e. Renal arteriography
2. A hyperdense renal cyst may also be termed a:
    a. probable malignancy.
    b. Bosniak II cyst.
    c. Bosniak III cyst.
    d. Bosniak IV cyst.
    e. probable angiomyolipoma.
3. The most generally accepted indication for core needle biopsy of a renal mass is a suspected clinical diagnosis of:

a. renal cell carcinoma (RCC).
b. renal oncocytoma.
c. renal cyst.
d. renal metastasis.
e. renal angiomyolipoma.

4. According to the American Urological Association (AUA) guidelines (Fig. 97.1), recommended postoperative radiographic surveillance of the chest after radical nephrectomy for T1N0M0 RCC is:
   a. no imaging studies.
   b. chest radiograph at 1 year.
   c. chest radiograph annually for 3 years.
   d. chest CT at 1 year and then chest radiograph annually for 2 years.
   e. chest radiograph annually for 5 years.
5. The European Organization for Research and Treatment of Cancer 30904 study randomly assigned patients to radical versus partial nephrectomy. Which of the following was an inclusion criterion?
   a. Clinical T1a tumor (<4.0 cm)
   b. Tumor size <5.0 cm
   c. Estimated glomerular filtration rate (GFR) >60 mL/min/1.73 $m^2$
   d. No hypertension
   e. Age <70 years
6. According to the AUA guidelines, following partial nephrectomy for pathologic stage T3aN0M0 RCC, it is recommended to perform surveillance abdominal CT scanning with what frequency?
   a. Never
   b. Every 6 months for at least 3 years and then annually to year 5
   c. Every year to year 5
   d. Every 2 years
   e. Every year for 2 years and then at year 5
7. Following partial nephrectomy of a solitary kidney, what is the most effective method of screening for hyperfiltration nephropathy?
   a. Urinary dipstick test for protein

b. Albumin-to-creatinine ratio
c. Iothalamate glomerular filtration rate (GFR) measurement
d. Serum creatinine measurement
e. Renal biopsy

8. The most accurate and practical assessment of renal function for routine use after nephrectomy is:
   a. serum creatinine measurement.
   b. urinary dipstick test for protein.
   c. 24-hours urinary protein measurement.
   d. iothalamate GFR measurement.
   e. serum creatinine–based estimation of GFR and analysis for proteinuria.
9. What is an important prerequisite for successful cryoablation of a renal tumor?
   a. Slow freezing
   b. Rapid thawing
   c. A single prolonged freeze-thaw cycle
   d. A double freeze-thaw cycle
   e. Freezing of tumor to a temperature of −10°C
10. Which two imaging modalities are the preferred and most accurate for demonstrating the presence and extent of an inferior vena caval tumor thrombus?
   a. Abdominal ultrasonography and CT
   b. MRI and renal artery angiography
   c. CT and MRI
   d. MRI and contrast venacavography
   e. Contrast venacavography and transesophageal ultrasonography
11. In patients undergoing complete surgical excision of RCC, the lowest 5-year survival rate is associated with which factor?
   a. Perinephric fat involvement
   b. Clear cell histology
   c. Subdiaphragmatic inferior vena caval involvement
   d. Intra-atrial tumor thrombus
   e. Lymph node involvement

12. A 45-year-old man has a 5-cm, exophytic RCC in the upper pole of a solitary left kidney and a single 2-cm left lower lung metastasis. What is the best treatment?
    a. Initial targeted therapy, then partial nephrectomy
    b. Partial nephrectomy, then targeted therapy

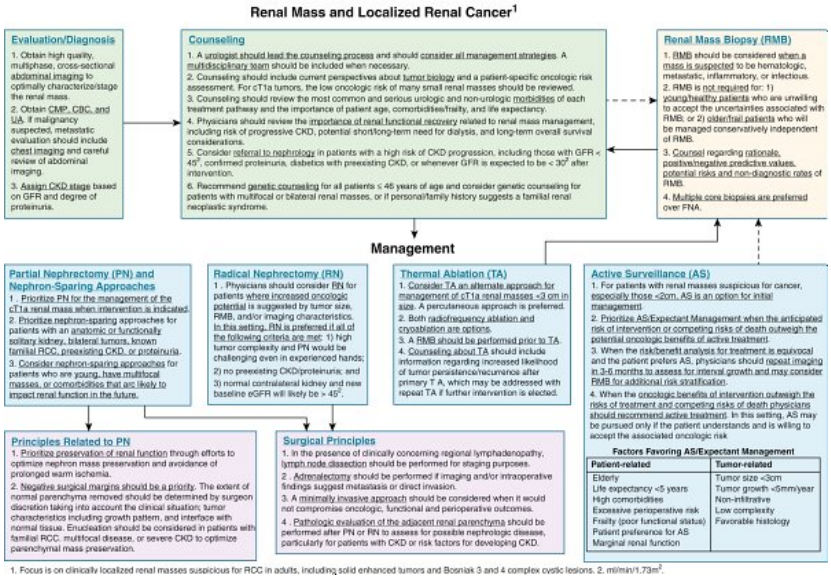



**FIG. 97.1**

    c. Staged partial nephrectomy and pulmonary lobectomy
    d. Simultaneous partial nephrectomy and pulmonary lobectomy
    e. Simultaneous radical nephrectomy and pulmonary lobectomy
13. A healthy 75-year-old man is referred after renal biopsy of a 3.0-cm centrally located renal mass. The biopsy is definitive for renal oncocytoma. The other kidney is normal, the serum creatinine level is 1.0 mg/dL, and there is no evidence of metastatic disease. What is the best next step?
    a. Open radical nephrectomy
    b. Laparoscopic nephroureterectomy
    c. Percutaneous thermal ablation (TA)
    d. Partial nephrectomy
    e. Observation with follow-up renal imaging
14. Which of the following agents demonstrated an oncologic benefit in the postoperative adjuvant setting for patients at high risk of

recurrence following nephrectomy?

a. High-dose interleukin-2 (IL-2)
b. Sunitinib
c. Autologous tumor vaccine
d. Pazopanib
e. Interferon-α (IFN-α)

15. Based on the 2017 AUA Guidelines, which of the following patients would not need to be considered for genetic counseling?
    a. A 45-year-old woman with a pT1a clear cell RCC.
    b. A 68-year-old man with a pT1a RCC with histology suggestive of succinate dehydrogenase (SDH) deficiency
    c. A 69-year-old man with 2 ipsilateral pT1a tumors, a 3.5-cm clear cell, and a 1.1-cm type 1 papillary
    d. A 62-year-old woman with a pT1b chromophobe RCC and moderate-sized lung cysts
    e. The 7-year-old son of a patient with von Hippel-Lindau (VHL) disease
16. Which two agents have the most similar mechanisms of action?
    a. Sunitinib and temsirolimus
    b. Pazopanib and axitinib
    c. Pembrolizumab and ipilimumab
    d. Nivolumab and cabozantinib
    e. Atezolizumab and ipilimumab
17. A 48-year-old woman with a history of seizure disorder presents with recurrent gross hematuria and left flank pain. Abdominal CT shows a large left perinephric hematoma associated with a 3.0-cm left renal angiomyolipoma. There are also multiple right renal angiomyolipomas ranging in size from 1.5 to 6.5 cm. What is the best management of the left renal lesion?
    a. Selective embolization
    b. Radical nephrectomy
    c. Observation
    d. Partial nephrectomy
    e. Laparoscopic exposure and renal cryoablative therapy
18. Which of the following statements is TRUE regarding cystic nephromas occurring in adults?

a. They are complex cystic lesions that are typically classified as Bosniak II to III.
b. They are malignant 2% to 5% of the time.
c. They are more common in men than in women.
d. When suspected, they should be treated by radical nephrectomy.
e. They are readily differentiated from cystic RCC on the basis of appropriate imaging studies.

19. Which environmental factor is most commonly accepted as a risk factor for RCC?
    a. Radiation therapy
    b. Antihypertensive medications
    c. Tobacco use
    d. Diuretics
    e. High-fat diet
20. Which of the following manifestations is common only in certain VHL families; specifically, which is common in those with type 2 and only rarely seen in type 1 VHL?
    a. RCC
    b. Pancreatic cysts or tumors
    c. Epididymal tumors
    d. Pheochromocytoma
    e. Inner ear tumors
21. Common manifestations of VHL disease include:
    a. RCC, pheochromocytoma, fibrofolliculoma
    b. angiomyolipoma, pheochromocytoma, fibrofolliculoma
    c. RCC, pheochromocytoma, hemangioblastoma
    d. angiomyolipoma, pheochromocytoma, hemangioblastoma
    e. RCC, thyroid carcinoma, leiomyoma
22. What is the most common cause of death in patients with the VHL syndrome?
    a. Renal failure
    b. Cerebellar hemangioblastoma
    c. Unrelated medical disease
    d. Pheochromocytoma
    e. RCC

23. The VHL syndrome tumor suppressor protein regulates the expression of which of the following mediators of biologic aggressiveness for RCC?
    a. Basic fibroblast growth factor
    b. Vascular endothelial cell growth factor
    c. Epidermal growth factor receptor (EGFR)
    d. Hepatocyte growth factor (scatter factor)
    e. P-glycoprotein (multiple drug resistance efflux protein)
24. What do the hereditary papillary RCC syndrome and VHL syndrome have in common?
    a. Mode of genetic transmission
    b. Chromosome 3 abnormalities
    c. Propensity toward tumor formation in multiple organ systems
    d. Inactivation of a tumor suppressor gene
    e. Nearly complete penetrance
25. Mutation of the met proto-oncogene in hereditary papillary RCC leads to:
    a. increased expression of hepatocyte growth factor.
    b. increased sensitivity to vascular endothelial growth factor (VEGF).
    c. inactivation of a tumor suppressor gene that regulates cellular proliferation.
    d. constitutive activation of the receptor for hepatocyte growth factor.
    e. increased expression of VEGF.
26. P-glycoprotein is a transmembrane protein that is involved in:
    a. immunotolerance.
    b. resistance to high-dose IL-2 therapy.
    c. resistance to cisplatin therapy.
    d. resistance to radiation therapy.
    e. efflux of large hydrophobic compounds, including many cytotoxic drugs.
27. Pathology demonstrates venous involvement limited to the main renal vein along with contralateral adrenal involvement with RCC. There is also a 6-cm bulky retroperitoneal lymph node replaced with cancer. What is the stage?

a. pT3aN1M0
b. pT3aN2M0
c. pT3aN1M1
d. pT3bN1M1
e. pT4N2M0

28. Which of the following is most likely to demonstrate an infiltrative growth pattern?
    a. Clear cell RCC
    b. Sarcomatoid variants of RCC
    c. Papillary RCC
    d. Chromophobe RCC
    e. Oncocytoma
29. What is the most common mutation identified in sporadic clear cell RCC?
    a. Activation of the met proto-oncogene
    b. Activation of the VHL gene
    c. Inactivation of the VHL gene
    d. Inactivation of p53
    e. Inactivation of genes on chromosome 9
30. Which of the following cytogenetic abnormalities is commonly associated with type 1 papillary RCC?
    a. Trisomy of chromosome 7
    b. Trisomy of the Y chromosome
    c. Loss of chromosome 17
    d. Loss of all or parts of chromosome 3
    e. Loss of chromosome 7
31. What percentage of RCCs are chromophobe cell carcinomas?
    a. 0% to 2%
    b. 3% to 5%
    c. 8% to 10%
    d. 12% to 15%
    e. 18% to 25%
32. Most renal medullary carcinomas are:
    a. found in patients with sickle cell disease.
    b. diagnosed in the fifth decade of life.
    c. responsive to high-dose chemotherapy.

d. genetically and histologically similar to papillary RCC.
e. metastatic at the time of diagnosis.

33. Which paraneoplastic syndrome associated with RCC can often be managed or palliated medically?
    a. Polycythemia
    b. Stauffer syndrome
    c. Neuropathy
    d. Hypercalcemia
    e. Cachexia
34. A healthy 64-year-old man is found to have a 6.0-cm solid, heterogeneous mass in the hilum of the right kidney. CT of the abdomen and pelvis shows interaortocaval lymph nodes enlarged to 2.5 cm. A chest radiograph and a bone scan are negative, and the contralateral kidney is normal. The serum creatinine level is 1.0 mg/dL. What is the best next step?
    a. Right radical nephrectomy and regional or extended lymph node dissection
    b. Abdominal exploration, sampling of the enlarged lymph nodes, and possible radical nephrectomy pending frozen section analysis
    c. CT-guided percutaneous biopsy of the lymph nodes
    d. CT-guided percutaneous biopsy of the tumor mass
    e. Systemic therapy followed by radical nephrectomy
35. Which of the following patients would be the best candidate for percutaneous biopsy or fine-needle aspiration of a renal mass?
    a. A 42-year-old man with a 2.5-cm Bosniak III complex renal cyst
    b. An 88-year-old man with a 1.7-cm solid, enhancing renal mass
    c. A 32-year-old woman with bilateral solid, enhancing renal masses ranging in size from 1.5 to 4.0 cm
    d. A 48-year-old woman with a 3.5-cm solid, enhancing renal mass with fat density present
    e. A 38-year-old woman with a fever, a urinary tract infection, and a 3.5-cm solid/cystic, enhancing renal mass

36. A 67-year-old man undergoes radical nephrectomy and inferior vena caval thrombectomy (level 2 tumor thrombus). The primary tumor is otherwise confined to the kidney, and the lymph nodes are not involved. What is the approximate 5-year cancer-free survival rate?
    a. 0% to 15%
    b. 16% to 30%
    c. 31% to 50%
    d. 51% to 70%
    e. 71% to 90%
37. Which of the following is NOT a predictor of cancer-specific survival after nephrectomy for RCC?
    a. Pathologic stage
    b. Tumor size
    c. Nuclear grade
    d. Patient age
    e. Histologic necrosis
38. Which of the following statements about renal lymphoma is TRUE?
    a. 5% to 10% of all lymphomas involving the kidney are primary tumors.
    b. The radiographic patterns manifested by renal lymphoma are diverse and can be difficult to differentiate from RCC.
    c. Percutaneous biopsy is rarely indicated if renal lymphoma is suspected.
    d. Renal failure associated with renal lymphoma is most often due to extensive parenchymal replacement by the malignancy.
    e. The most common pattern of renal involvement is by direct extension from adjacent retroperitoneal lymph nodes.
39. Which of the following would be considered diagnostic for renal angiomyolipoma (AML)?
    a. Hyperechoic pattern on ultrasonography
    b. Enhancement of >30 Hounsfield units on CT scan
    c. Small area measuring less than −20 Hounsfield units on non-enhanced CT
    d. Aneurysmal changes on renal arteriogram

e. Positive signal on T2 images of MRI

40. The main limitation of renal mass biopsy is:
    a. risk of needle tract seeding.
    b. difficulty differentiating the eosinophilic variants of RCC from renal oncocytoma.
    c. risk of pneumothorax.
    d. risk of hemorrhage.
    e. inability to determine subtype of RCC.
41. Which of the following tumors is most likely to be a malignant RCC?
    a. 2.5-cm hyperechoic complex cyst, with no enhancement with IV contrast
    b. 6.0-cm complex cyst with four thin septae
    c. 5.0-cm cyst with thin, curvilinear calcification
    d. 11-cm cyst with water density and homogeneous nature
    e. 3.0-cm solid lesion with fat associated with calcification
42. A common and pathogenic cytogenetic finding in children with RCC is:
    a. *VHL* mutation.
    b. *cMET* oncogene mutation.
    c. *p53* mutation.
    d. *TFE3* gene fusions.
    e. *PTEN* mutations.
43. According to the 2017 AUA Guidelines, which of the following statements about TA is TRUE?
    a. TA should be prioritized in elderly patients with comorbidities.
    b. In general, a tumor size threshold of 2.5 cm should be applied for decision making about TA.
    c. Renal mass biopsy should be performed at the time of TA whenever it might change management.
    d. Local recurrence is more common after TA than partial nephrectomy, but when repeat ablation is taken into account, similar local control rates are observed.
    e. Repeat ablation is feasible in some patients, but surgical resection is often required after failed TA.

44. What is the mechanism of action for nivolumab?
    a. Activates CTLA-4
    b. Blocks CTLA-4
    c. Activates PD-1
    d. Blocks PD-1
    e. Blocks PD-L1
45. According to the 8th edition of the AJCC TNM classification (2016), which of the following tumors would NOT be considered stage pT3a?
    a. 4.3 cm RCC with renal sinus fat invasion
    b. 12 cm RCC with involvement of the renal vein
    c. 5.6 cm RCC with invasion into the perinephric fat
    d. 4.8 cm RCC with invasion into the adrenal gland
    e. 6.4 cm RCC with collecting system involvement
46. Most tumors at various sites in the VHL syndrome share the following characteristic:
    a. Malignant behavior
    b. Hypervascularity
    c. Rapid growth rate
    d. High nuclear grade
    e. Symptomatic presentation
47. One major difference between hereditary papillary RCC syndrome and VHL syndrome is:
    a. pattern of genetic inheritance.
    b. age of onset.
    c. gender distribution.
    d. incidence of metastasis.
    e. incidence of associated tumors in nonrenal organ systems.
48. Which syndrome is most likely to exhibit aggressive behavior of RCC?
    a. VHL syndrome
    b. Hereditary papillary RCC syndrome
    c. Hereditary leiomyomatosis and RCC syndrome
    d. Birt-Hogg-Dubé (BHD) syndrome
    e. Familial oncocytosis

49. Spontaneous pneumothorax is occasionally observed in which of the following?
    a. VHL syndrome
    b. Hereditary papillary RCC syndrome
    c. Hereditary leiomyomatosis and RCC syndrome
    d. BHD syndrome
    e. Familial oncocytosis
50. Chromophobe RCC shares many characteristics with:
    a. oncocytoma.
    b. type 2 papillary RCC.
    c. clear cell RCC.
    d. mesoblastic nephroma.
    e. mixed epithelial and stromal tumor (MEST) of the kidney.
51. A finding that is diagnostic for collecting duct carcinoma is:
    a. central location and infiltrative growth pattern.
    b. aggressive clinical course.
    c. *p53* mutation.
    d. positive staining for *Ulex europaeus* lectin.
    e. sensitivity to chemotherapy.
52. Sarcomatoid differentiation is most commonly observed with which histologic subtypes of RCC?
    a. Clear cell and papillary
    b. Papillary and chromophobe
    c. Clear cell and collecting duct
    d. Clear cell and chromophobe
    e. Chromophobe and collecting duct
53. Which of the following factors has greatest utility for predicting bone metastasis from RCC?
    a. Tumor size
    b. Tumor grade
    c. Performance status (PS)
    d. Elevated alkaline phosphatase
    e. Invasion of the perinephric fat
54. The prognosis for a 3-cm tumor infiltrating the renal sinus fat is:
    a. similar to a pT1bN0 tumor.
    b. similar to a pT2aN0 tumor.

c. similar to a pT3a tumor with invasion of the perinephric fat laterally.
d. worse than a pT3a tumor with invasion of the perinephric fat laterally.
e. similar to a tumor with ipsilateral adrenal involvement.

55. The single most important prognostic factor for RCC is:
    a. tumor size.
    b. tumor grade.
    c. tumor stage.
    d. histologic subtype.
    e. PS.
56. The most accurate assessment of prognosis for patients with RCC is usually provided by:
    a. tumor size.
    b. clinician judgment.
    c. tumor stage.
    d. integrated analysis of prognostic factors.
    e. PS.
57. Which of the following indicates a tumor that is NOT correctly staged according to the 2016 TNM staging system for RCC?
    a. Localized RCC, 8.5 cm: *pT2a*
    b. RCC with direct ipsilateral adrenal involvement: *pT3a*
    c. RCC with metastatic involvement of the adrenal gland: *pM1*
    d. RCC with direct invasion of the pelvicalyceal system: *pT3a*
    e. RCC with three lymph node metastases: *pN1*
    a. 2.5-cm centrally located tumor after radiofrequency ablation
58. What is the most common form of renal sarcoma?
    a. Liposarcoma
    b. Rhabdomyosarcoma
    c. Fibrosarcoma
    d. Leiomyosarcoma
    e. Angiosarcoma
59. The most useful prognostic factors for renal sarcoma are:
    a. tumor size and grade.
    b. tumor stage and grade.
    c. histologic subtype and stage.

d. tumor stage and ploidy status.
e. margin status and grade.

60. Which of the following renal tumors has the best prognosis?
    a. Sarcoma
    b. Carcinoid
    c. Adult Wilms
    d. Primitive neuroectodermal tumor
    e. Small cell
61. Patients with which RCC subtype are most likely to benefit from targeted therapy, such as tyrosine kinase inhibitors (TKIs)?
    a. Papillary RCC
    b. Clear cell RCC
    c. Renal medullary carcinoma
    d. Collecting duct carcinoma
    e. Chromophobe RCC
62. Agents targeting which of the following signaling pathways in clear cell RCC have significant antitumor effects in patients with metastatic disease?
    a. VEGF and EGFR
    b. *p53* and EGFR
    c. VEGF and mammalian target of rapamycin (mTOR)
    d. mTOR and transforming growth factor-α (TGF-α)
    e. All of the above
63. Of the following, which is the greatest determinant of renal function after partial nephrectomy?
    a. Surgical approach (open vs. minimally invasive)
    b. Tumor size
    c. Absence of a functioning contralateral kidney
    d. Renal function before partial nephrectomy
    e. Gender
64. Which of the following statements is TRUE regarding chronic kidney disease (CKD)?
    a. CKD due to surgery has the same impact as CKD due to medical causes.
    b. Serum creatinine below 1.4 mg/dL excludes the possibility of CKD.

c. CKD can be diagnosed based on a single estimated GFR value less than 60 mL/min/1.73 $m^{2.}$
d. Increasing CKD stage has been associated with an increase in morbid cardiovascular events for subjects in the general population.
e. Choice of intervention for localized renal malignancy has little impact on development or progression of CKD.

65. Which of the following is an indication for adrenalectomy at the time of partial nephrectomy?
    a. 6-cm upper pole renal tumor
    b. 4-cm adrenal lesion measuring −20 Hounsfield units on noncontrast CT scan
    c. Bilateral adrenal hyperplasia
    d. 3-cm renal tumor adjacent to the adrenal gland on CT scan, but readily separable from the adrenal gland at surgery
    e. 1.5-cm adrenal lesion that is bright on T2-weighted MRI
66. Which of the following statements is FALSE regarding molecular imaging for renal tumors?
    a. $^{99m}$Tc-sestamibi SPECT/CT is based on uptake by mitochondria, which are in abundance in oncocytomas and hybrid oncocytic/chromophobe tumors.
    b. $^{124}$I-girentuximab targets carbonic anhydrase IX present on clear cell RCC.
    c. Fluorodeoxyglucose positron emission tomography (FDG-PET)-CT is useful for staging RCC.
    d. Prostate-specific membrane antigen (PSMA) imaging can identify clear cell RCC.
    e. While early studies are promising, validation studies are needed before widespread acceptance of current molecular imaging techniques.
67. Which of the following statements most accurately characterizes the role of lymphadenectomy for RCC?
    a. Offers a staging and therapeutic benefit for patients with clinically localized RCC

b. Offers staging information but not a therapeutic benefit for all patients with RCC
c. Offers a therapeutic benefit mostly for patients with locally advanced RCC
d. Compliance with well-described templates is important for achieving a therapeutic benefit
e. Demonstrated a small but statistically significant survival advantage in a randomized trial

68. According to the 2017 AUA Guidelines, which of the following patients with kidney cancer will not need to be considered for nephrology consultation?
    a. A patient with 50/50 split renal function and a preoperative GFR of 70 mL/min/1.73 $m^2$ who will be undergoing radical nephrectomy
    b. A patient with preoperative GFR of 42 mL/min/1.73 $m^2$
    c. A patient with confirmed proteinuria
    d. A diabetic with preoperative GFR <60 mL/min/1.73 $m^2$
    e. A patient with a solitary kidney, a preoperative GFR of 36 mL/min/1.73 $m^2$, and a R.E.N.A.L. 10 lesion

## Pathology

1. A 49-year-old man has a biopsy of a peripheral lower pole exophytic 4-cm mass of the right kidney. The histology is depicted in Fig. 97.2 and is read as clear cell carcinoma. Metastatic workup is negative. He should be advised to:
    a. receive targeted chemotherapy.
    b. undergo radical nephrectomy.
    c. have the tumor ablated with cryotherapy.
    d. undergo partial nephrectomy.
    e. receive radiation therapy.
2. A 48-year-old woman has a right radical nephrectomy for a 6-cm mass. Preoperative metastatic workup was negative. The pathology is illustrated in Fig. 97.3 and is read as collecting duct carcinoma. She should be advised to:

a. receive targeted chemotherapy.
b. receive radiation therapy to the nephrectomy bed.
c. receive platinum-based chemotherapy.
d. follow up with her primary care physician.
e. be followed closely because the development of metastatic disease is likely.

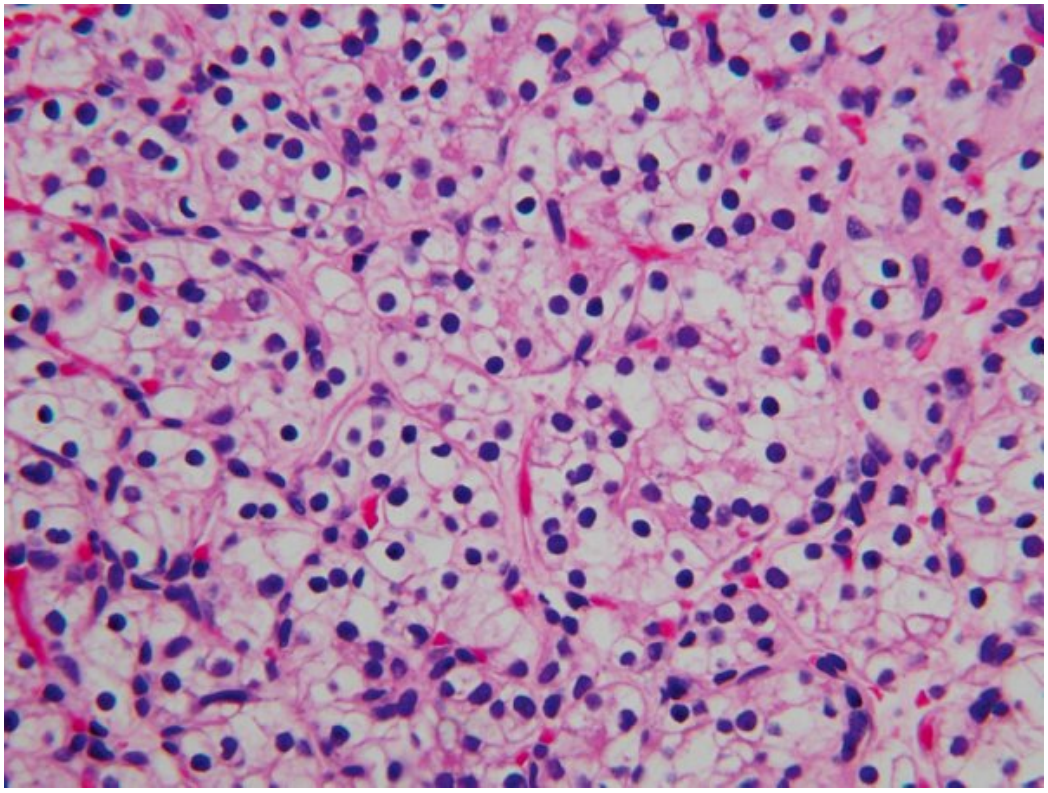


**FIG. 97.2**
From Bostwick DG, Cheng L: *Urologic surgical pathology*, 2nd ed, Edinburgh, 2008, Mosby.

## Imaging

1. See Fig. 97.4. A 55-year-old man with hematuria has this contrast-enhanced CT scan for evaluation. The most appropriate therapy is:
   a. laparoscopic nephron-sparing surgery.

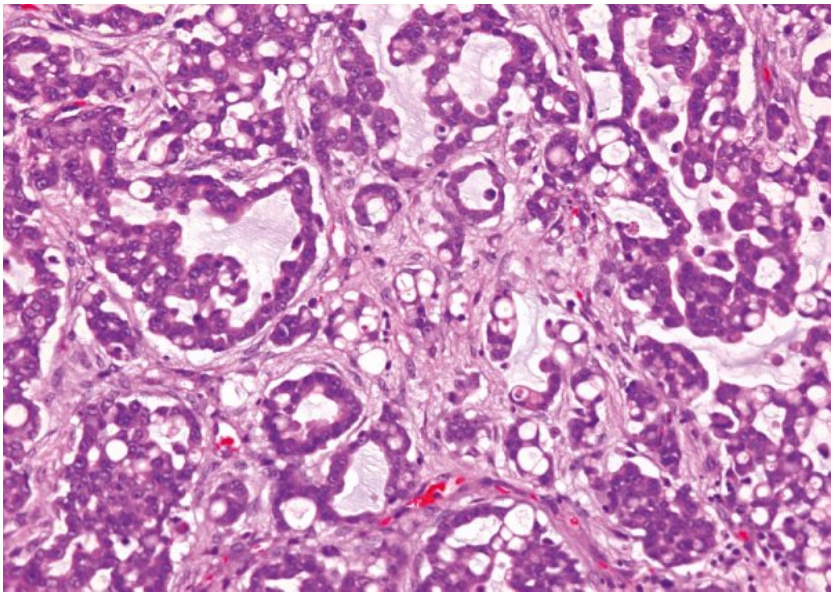
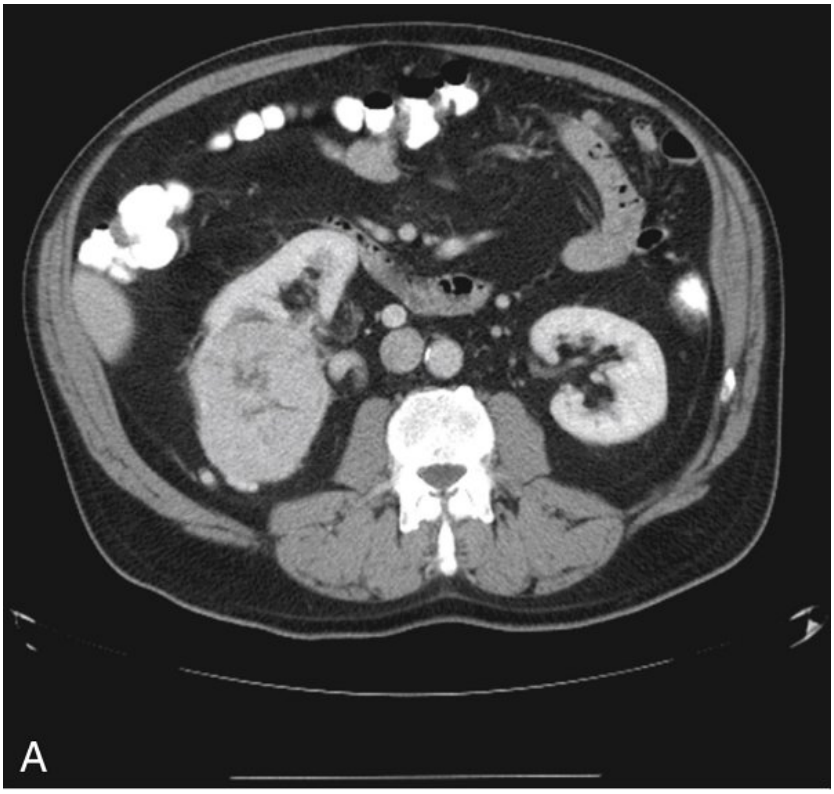


**FIG. 97.3**

From Bostwick DG, Cheng L. *Urologic surgical pathology,* 2nd ed, Edinburgh, 2008, Mosby.


A

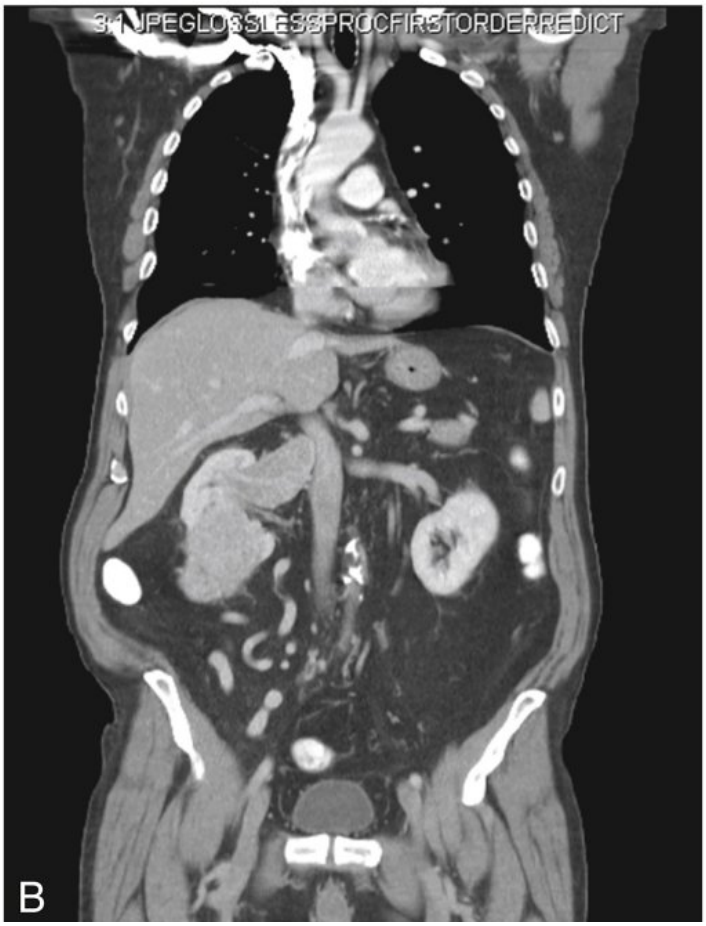

B

FIG. 97.4

b. radical nephrectomy.
c. open nephron-sparing surgery.
d. radiofrequency ablation.
e. cryoablation.

2. See Fig. 97.5. A 45-year-old man with no urinary symptoms has this axial contrast-enhanced CT scan. What is the most likely diagnosis?
   a. Bosniak II-F lesion
   b. Bosniak IV—cystic RCC
   c. Bosniak II cyst
   d. Bosniak III cyst
   e. Bosniak I cyst

## Answers

1. **c. Computed tomography (CT) with and without contrast enhancement.** A dedicated triple-phase renal CT scan remains the single most important radiographic image to delineate the nature of a renal mass. In general, any renal mass that enhances with administration of intravenous contrast material on CT scanning should be considered a RCC until proved otherwise.
2. **b. Bosniak II cyst.** Category II lesions are minimally complicated cysts that are benign but have some radiologic findings that cause concern. Classic hyperdense renal cysts are small (<3 cm), round, sharply marginated, and do not enhance after administration of contrast material. Hyperdense cysts that are 3 cm or larger are classified as *Bosniak IIF* lesions.
3. **d. Renal metastasis.** The traditionally accepted indications for biopsy of a renal mass are when a renal abscess or infected cyst is suspected, or when differentiating RCC from metastatic malignancy or renal lymphoma. Core needle biopsy is now performed with increased frequency for the evaluation of renal

masses in other circumstances, particularly for patients in whom a wide variety of treatment options are under consideration.

4. **c. Chest radiograph annually for 3 years.** Surveillance for recurrent malignancy after radical nephrectomy for RCC can be tailored according to the initial pathologic tumor stage. This patient is low risk (pT1N0M0), and the AUA guidelines recommend an annual chest radiograph for 3 years and only as clinically indicated beyond that time period.
5. **b. Tumor size <5.0 cm.** A solitary tumor and a normal contralateral kidney were also required, but criteria for the latter were not well defined, at least based on current perspectives about functional assessment that depend on eGFR rather than serum creatinine level.
6. **b. Every 6 months for at least 3 years and then annually to year 5.** Surveillance for recurrent malignancy after nephron-sparing surgery for RCC can be tailored according to the initial pathologic tumor stage. This patient is intermediate to high risk, and the AUA Guidelines recommend a baseline abdominal scan (CT or MRI) within 3 to 6 months following surgery with continued imaging (ultrasonography, CT, or MRI) every 6 months for at least 3 years and annually thereafter to year 5. Imaging beyond 5 years may be performed at the discretion of the clinician.



FIG. 97.5

7. **b. Albumin-to-creatinine ratio.** Patients who undergo nephron-sparing surgery for RCC may be left with a relatively small amount of renal tissue. These patients are at risk for long-term renal functional impairment from hyperfiltration renal injury. Because proteinuria is the initial manifestation of the phenomenon, an albumin-to-creatinine ratio should be obtained yearly in patients with a solitary remnant kidney to screen for hyperfiltration nephropathy. Traditionally a 24-hour urine protein was obtained on an annual basis, but this is more burdensome, and for screening purposes the albumin-to-creatinine ratio is more suitable. If positive, more rigorous testing, such as the 24-hour urine protein, can be considered to more accurately quantify the degree of proteinuria.
8. **e. Serum creatinine–based estimation of GFR and analysis for proteinuria. Identification and classification of CKD is best performed by assessing cause, estimated GFR (eGFR), and extent of albuminuria.** At present, there are several formulas in clinical use, including the MDRD and CKD-EPI formulas, each of which is an improvement over using serum creatinine alone for identification of patients with or at risk for CKD. Serum levels of

creatinine are dependent on gender, muscle mass, and other factors and can therefore lead to an underestimation of kidney disease in certain populations, such as thin, elderly women. Urinary creatinine measurement is impractical and provides only marginally more valuable information than serum creatinine. Urinary protein measurement can identify patients with early signs of kidney disease (proteinuria), and there are multiple methods to assess this, including 24-hour urinary protein and spot urine studies (albumin-to-creatinine ratio, protein-to-creatinine ratio); none has been shown to be better than GFR estimation as a screening test. Direct measurement of GFR using iothalamate (or other agents) is costly and not routinely available; it is therefore impractical in most settings.

9. **d. A double freeze-thaw cycle.** Renal cryosurgery is an ablative nephron-sparing treatment option for RCC that can be performed percutaneously under radiographic guidance or laparoscopically under direct vision and ultrasound guidance. The aim of cryosurgery is to ablate the same predetermined volume of tissue that would have been removed had a conventional surgical excision been performed. Established critical prerequisites for successful cryosurgery include rapid freezing, gradual thawing, and a repetition of the freeze-thaw cycle.
10. **c. CT and MRI.** Both CT and MRI are noninvasive and accurate modalities for demonstrating both the presence and the distal extent of vena caval involvement. Although MRI has been recommended as the test of choice at most centers, many studies have demonstrated that a properly performed multiplanar CT also provides sufficient information for surgical planning, and it has become the preferred diagnostic study at many centers.
11. **e. Lymph node involvement.** In most studies, the presence of lymph node or distant metastases has carried a dismal prognosis that is much more pronounced than the other distractors.
12. **d. Simultaneous partial nephrectomy and pulmonary lobectomy.** The subset of patients with metastatic RCC and a solitary metastasis, estimated at between 1.6% and 3.2% of patients, may benefit from nephrectomy with resection of the

metastatic lesion. This patient also needs partial nephrectomy to preclude the need for dialysis. Of note, both procedures could be performed with a minimally invasive approach under the same anesthesia.

13. **e. Observation with follow-up renal imaging.** Renal mass biopsy is now performed with increased frequency and should be considered in an elderly patient such as this. For those in whom nonextirpative options are being considered, biopsy can provide important information, such as a definitive nonmalignant diagnosis (as in this example). Given the benign nature of renal oncocytomas, the best answer is observation with follow-up imaging at an interval between 6 and 12 months, if clinically warranted based on the patient's overall health and life expectancy.
14. **b. Sunitinib. Sunitinib** demonstrated a significantly longer disease-free survival compared to placebo (6.8 vs. 5.6 years) in the phase 3 S-TRAC trial that randomized 615 men with pT2 (high-grade), pT3-4, or N1 clear cell RCC to sunitinib or placebo. Importantly, patients receiving therapy experienced significantly higher rates of toxic events and there was no difference in overall survival. All other therapies listed failed to demonstrate an oncologic benefit in the adjuvant setting.
15. **c. A 69-year-old man with 2 ipsilateral pT1a tumors, a 3.5-cm clear cell, and a 1.1-cm type 1 papillary. A 69-year-old man with 2 ipsilateral pT1a tumors, a 3.5-cm clear cell and a 1.1-cm type 1 papillary probably does not need to consider genetic counseling, unless the family or personal history is otherwise suggestive of a familial syndrome.** Genetic counseling is recommended for all patients with kidney cancer who are 46 years of age or younger. When histology suggests possible SDH deficiency, further testing should be pursued, and a history of chromophobe RCC and lung cysts or a prior spontaneous pneumothorax is suggestive for BHD syndrome. Relatives of patients with known familial RCC should be strongly considered for genetic counseling, and surveillance ideally will be initiated at an early age. An elderly patient with multifocal RCC may not need

to be considered for genetic counseling, particularly if there is discordance in the histologies.

16. **b. Pazopanib and axitinib. Pazopanib and axitinib both target the VEGF pathway, and specifically the VEGF receptor, and are TKI.** Sunitinib and cabozantinib are also TKIs, and temsirolimus is an mTOR inhibitor. Pembrolizumab, atezolizumab, ipilimumab, and nivolumab are all checkpoint inhibitors (Fig. 97.6). Pembrolizumab and nivolumab block PD-1, while atezolizumab blocks PD-L1 and ipilimumab blocks CTLA-4.
17. **a. Selective embolization.** Most patients with acute or potentially life-threatening hemorrhage will require total nephrectomy if exploration is performed, and if the patient has TSC, bilateral disease, preexisting renal insufficiency, or other medical or urologic disease that could affect renal function in the future, selective embolization should be considered. In such circumstances, selective embolization can temporize by controlling hemorrhage and in many cases will prove to be definitive treatment. This patient almost certainly has TSC given bilateral AMLs and history of seizures.
18. **a. They are complex cystic lesions that are typically classified as Bosniak II to III. Cystic nephromas are benign renal neoplasms that occur most commonly in middle-aged women.** They appear to be genetically related to MEST but generally have a somewhat different radiographic appearance. Unlike MEST, which contain a solid stromal component and often appear as solid or Bosniak IV lesions on cross-sectional imaging, cystic nephromas are typically characterized as complex cystic lesions without a solid component.
19. **c. Tobacco use.** The most generally accepted environmental risk factor for RCC is tobacco use, although the relative associated risks have been modest, ranging from 1.4 to 2.3 when compared with controls. All forms of tobacco use have been implicated, with risk increasing with cumulative dose or pack-years. **Other well-established risk factors include obesity and hypertension.**
20. **d. Pheochromocytoma.** The familial form of clear cell RCC is the VHL syndrome. Major manifestations include the development of RCC, pheochromocytoma, retinal angiomas, and

hemangioblastomas of the brainstem, cerebellum, or spinal cord. **Penetrance for all of these traits is far from complete, and some, such as pheochromocytomas, tend to be clustered in certain families but not in others.**

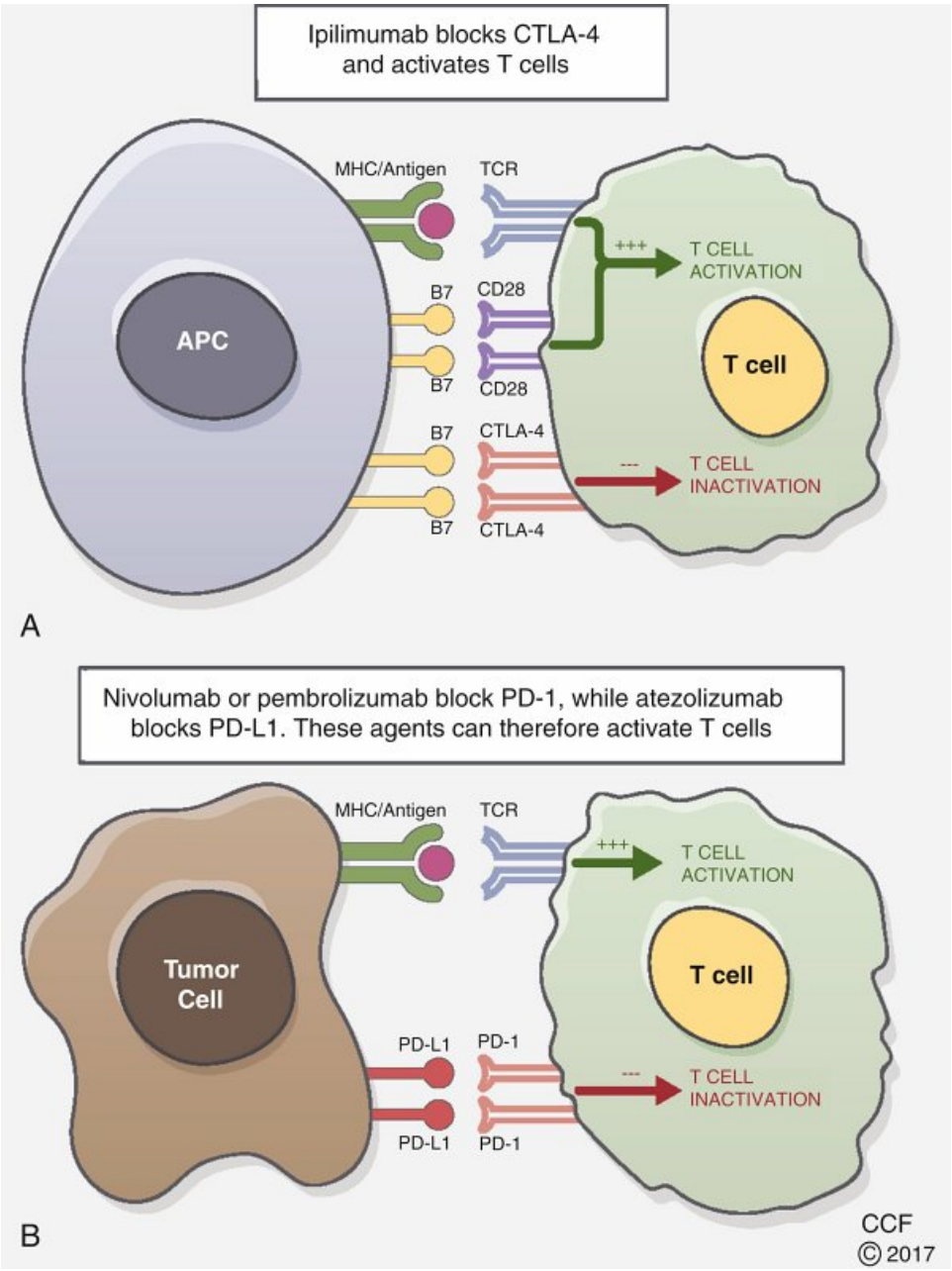



FIG. 97.6

21. **c. RCC, pheochromocytoma, hemangioblastoma.** These are the three most common tumor types occurring in VHL disease. Facial fibrofolliculomas occur in BHD syndrome, and uterine leiomyomas are commonly seen in hereditary leiomyomatosis and RCC syndrome.
22. **e. RCC.** With improved management of the central nervous system manifestations of the disease, RCC is the most common cause of mortality in patients with VHL syndrome.
23. **b. Vascular endothelial cell growth factor.** Inactivation or mutation of the VHL gene leads to dysregulated expression of hypoxia inducible factor-1, an intracellular protein that plays an important role in regulating cellular responses to hypoxia, starvation, and other stresses. This in turn leads to a several-fold upregulation of the expression of VEGF, the primary proangiogenic growth factor in RCC, contributing to the pronounced neovascularity associated with this carcinoma.
24. **a. Mode of genetic transmission.** Studies of families with hereditary papillary renal carcinoma (HPRC) have demonstrated an autosomal dominant mode of transmission, similar to VHL disease. VHL is caused by inactivation or mutation of a tumor suppressor gene, whereas HPRC is caused by activation of an oncogene.
25. **d. Constitutive activation of the receptor for hepatocyte growth factor.** Missense mutations of the *met* proto-oncogene at 7q31 were found to segregate with the disease, implicating it as the relevant genetic locus. The protein product of this gene is the receptor tyrosine kinase for the hepatocyte growth factor, which plays an important role in regulating the proliferation and differentiation of epithelial and endothelial cells in a wide variety of organs, including the kidney. Most of the mutations in hereditary papillary RCC have been found in the tyrosine kinase domain of *met* and lead to constitutive activation.
26. **e. Efflux of large hydrophobic compounds, including many cytotoxic drugs.** P-glycoprotein is a 170-kDa transmembrane protein expressed by 80% to 90% of RCCs that acts as an energy-dependent efflux pump for a wide variety of large hydrophobic compounds, including several cytotoxic drugs.

27. **c. pT3aN1M1. Isolated renal vein involvement is now classified as T3a, and nodal classification has been simplified such that all nodal involvement is now classified as N1.** Contiguous invasion of the ipsilateral adrenal is now classified as T4, but metastatic involvement of the contralateral (or ipsilateral) adrenal gland is classified as M1, reflecting a likely hematogenous pattern of dissemination.
28. **b. Sarcomatoid variants of RCC.** Most RCCs are round to ovoid and circumscribed by a pseudocapsule of compressed parenchyma and fibrous tissue rather than a true histologic capsule. Unlike upper tract transitional cell carcinomas (TCCs), most RCCs are not grossly infiltrative, with the notable exception of some sarcomatoid variants.
29. **c. Inactivation of the VHL gene.** Chromosome 3 alterations and VHL mutations are common in conventional RCC, and mutation or inactivation of this gene has been found in over 75% of sporadic cases.
30. **a. Trisomy of chromosome 7.** The cytogenetic abnormalities associated with type 1 papillary RCC are characteristic and include trisomy of chromosomes 7 and 17.
31. **b. 3% to 5%. Chromophobe cell carcinoma is a distinctive histologic subtype of RCC that appears to be derived from the cortical portion of the collecting duct.** It represents 3% to 5% of all RCCs.
32. **e. Metastatic at the time of diagnosis. Renal medullary carcinoma is a rare histologic subtype of RCC that occurs almost exclusively in association with sickle cell trait.** It is typically diagnosed in young African Americans, often in the third decade of life. Many cases are both locally advanced and metastatic at the time of diagnosis. Most patients have not responded to therapy and have succumbed to their disease in a few to several months.
33. **d. Hypercalcemia. Hypercalcemia has been reported in up to 13% of patients with RCC and can be due to either paraneoplastic phenomena or osteolytic metastatic involvement of the bone. The production of parathyroid hormonelike**

**peptides is the most common paraneoplastic etiology, although tumor-derived 1,25-dihydroxyvitamin $D_3$ and prostaglandins may contribute in a minority of cases.** Medical management includes vigorous hydration followed by diuresis with furosemide and the selective use of bisphosphonates, corticosteroids, and/or calcitonin.

34. **a. Right radical nephrectomy and regional or extended lymph node dissection.** An aggressive surgical approach is still preferred because it will likely prolong survival and represents the only realistic chance for a cure. Lymph nodes in this size range are most likely malignant, and this patient will likely need to consider adjuvant clinical trials. An extended lymph node dissection includes the interaortocaval nodes and nodes alongside and behind the ipsilateral great vessel from the crus of the diaphragm to the ipsilateral common iliac artery.
35. **e. A 38-year-old woman with a fever, urinary tract infection, and a 3.5-cm solid/cystic, enhancing renal mass.** Patients with flank pain, a febrile urinary tract infection, and a renal mass should be offered percutaneous biopsy and/or aspiration to establish a diagnosis of renal abscess rather than malignancy, and to provide drainage if abscess is confirmed.
36. **c. 31% to 50%.** Venous involvement with tumor thrombi was once thought to be a dismal sign for patients with RCC, but more recent studies suggest that many patients with tumor thrombi can be salvaged with an aggressive surgical approach. These studies document 31% to 50% 5-year survival rates for patients with IVC thrombus thrombi, as long as the tumor is otherwise confined to the kidney.
37. **d. Patient age.** Although patient age and comorbidity are important predictors of overall survival in patients with RCC and strongly affect the choice of treatment in these patients, they have no effect on the likelihood of dying of cancer-specific causes. Each of the other factors is a predictor of cancer-specific mortality from RCC and has been incorporated into one or more RCC prognostic algorithms for cancer-specific outcomes.

38. **b. The radiographic patterns manifested by renal lymphoma are diverse and can be difficult to differentiate from RCC.** Five different radiographic patterns have been described for lymphoma involving the kidney, including a solitary mass, occasionally making it difficult to differentiate from RCC.
39. **c. Small area measuring less than −20 Hounsfield units on non-enhanced CT. The presence of even a small focus of fat, as evidenced by a density less than − 20 HU on a non-enhanced CT scan, is diagnostic for AML.** The findings described in a, b, and d are all suggestive of, but not diagnostic for, renal AML.
40. **b. Difficulty differentiating the eosinophilic variants of RCC from renal oncocytoma.** The main limitation of renal mass biopsy is difficulty differentiating renal oncocytoma, the most common benign renal mass, from eosinophilic variants of conventional, papillary, and chromophobe RCC on biopsy material. While the non-diagnostic rate is between 10% and 15%, the risk of complications is low in the modern era with the use of smaller-gauge needles, and needle tract seeding with RCC appears to be a rare event.
41. **e. 3.0-cm solid lesion with fat associated with calcification.** Tumors with calcification associated with fat are uncommon but are almost always malignant RCC. In this setting the fat is thought to be a reactive process related to tumor necrosis. Calcification is virtually never seen in association with AML. The lesions described in a–c are Bosniak II renal cysts, with risk of malignancy of <10%. The lesion described in d is a simple cyst and highly likely to be benign despite its large size.
42. **d. *TFE3* gene fusions.** Mutations or translocations resulting in *TFE3* gene fusions are common in RCC occurring in the pediatric population. Although these cancers often present with advanced stage, the t(X;17) variant frequently follows an indolent course, while t(X;1) cancers can recur with late lymph node metastases.
43. **d. Local recurrence is more common after TA than partial nephrectomy, but when repeat ablation is taken into account, similar local control rates are observed.** Renal mass biopsy should always be performed prior to or concomitant with TA, as it

will help guide surveillance. Most local failures of TA can be salvaged with repeat TA, and only a small minority of such patients end up requiring salvage surgery. Success rates TA, for tumors <3.0 cm have been encouraging, and this threshold is advocated in the Guidelines as a general reference point. TA can be a good choice in some elderly patients with comorbidities, but active surveillance is also a reasonable option and in reality is often a better choice for many such patients.

44. **c. Activates PD-1. The mechanism of action of nivolumab is to block PD-1 and thus activate the immune system.** Pembrolizumab also blocks PD-1, while atezolizumab blocks PD-L1 and ipilimumab blocks CTLA-4.
45. **d. 4.8 cm RCC with invasion into the adrenal gland.** Direct invasion of the adrenal gland is pT4 and metastatic involvement of the adrenal gland is pM1. All of the other characteristics are indicative of pT3a tumors, including collecting system involvement (a modification made in the 8th edition).
46. **b. Hypervascularity.** Tumors in the VHL syndrome include adrenal pheochromocytoma, retinal angiomas, cerebellar and brainstem hemangioblastoma, RCC, and others. Most are relatively slow growing and asymptomatic if patients are evaluated and screened in a proactive manner. The common feature is that almost all are hypervascular.
47. **e. Incidence of associated tumors in nonrenal organ systems.** The incidence of nonrenal tumors is low in the hereditary papillary RCC syndrome in contrast to VHL syndrome, in which patients commonly develop tumors in the eyes, spinal cord, cerebellum, adrenal glands, inner ear, epididymis, and pancreas.
48. **c. Hereditary leiomyomatosis and RCC syndrome.** Malignant behavior is particularly common in the hereditary leiomyomatosis and RCC syndrome, and proactive and aggressive surgical management is recommended.
49. **d. BHD syndrome.** Lung cysts and spontaneous pneumothoraces are well described and relatively common findings in the BHD syndrome.

50. **a. Oncocytoma.** Both chromophobe RCC and renal oncocytoma are derived from the distal tubules, and both are commonly observed in the BHD syndrome. There are also some overlapping cytogenetic changes, all suggesting a potential relationship between these renal tumors.
51. **d. Positive staining for *Ulex europaeus* lectin.** *Ulex europaeus* lectin is expressed by the normal collecting duct, and tumor staining suggests origin from this structure. Most collecting duct carcinomas are centrally located and exhibit an infiltrative growth pattern and aggressive clinical course, but this is also true for poorly differentiated TCCs of the renal pelvis or centrally located sarcomatoid RCC. Collecting duct RCC is not typically responsive to chemotherapy.
52. **d. Clear cell and chromophobe.** Sarcomatoid differentiation is most commonly found in association with clear cell and chromophobe RCC.
53. **c. Performance status (PS).** Poor PS can be used to segregate patients when deciding whether to obtain a bone scan for metastatic RCC. Shvarts and colleagues (2004) [a] have shown that patients with good PS (ECOG PS = 0), no evidence of extraosseous metastases, and no bone pain were extremely low risk for bone metastasis and did not benefit from bone scanning. They recommended a bone scan for all other patients, and the incidence of bone metastasis in this group was >15%.
54. **d. Worse than a pT3a tumor with invasion of the perineph-ric fat laterally.** Invasion of the perisinus fat medially has been shown to be a poor prognostic sign. Medial invasion places the tumor in proximity to the venous system and likely increases the risk of metastatic dissemination. Ipsilateral adrenal involvement is even worse, and these are now classified as pT4 if due to direct local extension, or pM1 otherwise, consistent with a hematogenous route of dissemination.
55. **c. Tumor stage.** Although not truly a single factor because it combines tumor size with other information obtained from final pathologic analysis, tumor stage is the most powerful individual

predictor of oncologic outcomes. When incorporated into a multi-predictor analysis, such as a nomogram or other multivariable analysis, the predictive ability increases further.

56. **d. Integrated analysis of prognostic factors.** Integrated analysis of a variety of factors such as tumor size, stage, and grade, PS, and histologic subtype has yielded the most accurate prognostication for RCC. Several studies have documented that nomograms and other algorithms outperform traditional staging systems, clinical opinion, individual risk factors, and chance.
57. **b. RCC with direct ipsilateral adrenal involvement: *pT3a.*** Several studies have demonstrated that RCC directly invading the adrenal gland is associated with poorer prognosis than RCC with perinephric or renal sinus fat invasion. Direct ipsilateral adrenal involvement is now grouped with other RCCs that extend beyond Gerota fascia as pathologic stage T4 *(pT4).* These patients have a high risk for disease recurrence or progression.
58. **d. Leiomyosarcoma.** Leiomyosarcoma is the most common histologic subtype of renal sarcoma, accounting for 50% to 60% of such tumors. The most common type of sarcoma in the retroperitoneum is liposarcoma.
59. **e. Margin status and grade.** Margin status and tumor grade are the primary prognostic factors for sarcoma. Patients with high-grade disease are at risk for systemic metastasis, and those with low-grade disease are at risk for local recurrence. Wide local excision with negative margins is essential for minimizing the risk of recurrence for sarcomas because these tumors are derived from the mesenchymal tissues and are typically infiltrative, and thus do not respect natural barriers.
60. **b. Carcinoid.** All of these tumor types have a relatively poor prognosis except for renal carcinoid, which tends to be associated with good outcomes in most patients.
61. **b. Clear cell RCC.** Targeted molecular therapies, such as TKI, mostly target VEGF and thus the pathways that are typically upregulated in clear cell RCC. Most objective responses have been observed in individuals with clear cell RCC. Standard of care for

the treatment of metastatic non–clear cell RCC is not well defined at present.

62. **c. VEGF and mammalian target of rapamycin (mTOR).** Several clinical trials indicate that agents targeting VEGF signaling, including sunitinib, sorafenib, axitinib, pazopanib, levantanib, caboxantinib, and mTOR pathway signaling, including temsirolimus and everolimus, demonstrate substantial tumor responses or significant improvement in progression-free or overall survival.
63. **d. Renal function before partial nephrectomy.** When compared with radical nephrectomy, partial nephrectomy has been associated with better renal functional outcomes. The main determinants of renal function after partial nephrectomy are the quality of the kidney prior to surgery and the quantity of vascularized parenchymal mass that is preserved after excision of the tumor and reconstruction of the kidney. Warm ischemia time, if prolonged, can also contribute to a decline in renal function after partial nephrectomy.
64. **d. Increasing CKD stage has been associated with an increase in morbid cardiovascular events, hospitalization, and death on a longitudinal basis for subjects in the general population.** In this setting, CKD is predominantly due to medical causes. CKD due to surgical causes appears to be more stable and less deleterious than that due to medical causes.
65. **e. 1.5-cm adrenal lesion that is bright on T2-weighted MR imaging.** Several pathologic adrenal lesions can be diagnosed based on their radiographic characteristics without histologic confirmation. Lesions that are bright on T2-weighted MR imaging are suspicious for pheochromocytomas and should be surgically removed. Careful preoperative and intraoperative management are essential for safe management in this circumstance.
66. **c. Fluorodeoxyglucose positron emission tomography (FDG-PET)-CT is useful for staging RCC. FDG-PET-CT is not useful for the staging of RCC** due to high levels of visual conspicuity in the kidney and at the most common sites of metastatic disease. Sestamibi is taken up by mitochondria, which are in abundance in

oncocytolmas and hybrid oncocytic/chromophobe tumors, and a strong signal is suggestive of these histologies. $^{124}$I-girentuximab, which targets carbonic anhydrase IX primarily present on clear cell RCC, and PSMA (epithelial membrane target) remain in development for clear cell RCC.

67. **b. Offers staging information but not a therapeutic benefit for all patients with RCC.** Lymphadenectomy may be performed in patients with clinically suspicious (radiographic or intraoperative) lymphadenopathy for staging purposes and, due to lack of data indicating a reliable therapeutic benefit, need not be performed routinely in patients with localized kidney cancer and clinically negative nodes. The EORTC 30881 randomized trial of lymphadenectomy at RN failed to show a survival advantage for patients undergoing lymphadenectomy.
68. **a. A patient with 50/50 split renal function and a preoperative GFR of 70 mL/min/1.73 m$^2$ who will be undergoing radical nephrectomy. A patient with 50/50 split renal function and a preoperative GFR of 70 mL/min/1.73m $^2$ who will be undergoing radical nephrectomy will likely end up with a new baseline GFR of about 40-42 mL/min/1.73m $^2$ and thus will likely not need referral to nephrology based on the 2017 AUA Guidelines for Renal Mass and Localized RCC.** Per the guidelines, referral to nephrology should be considered in patients at high risk for GFR progression, including those with preoperative GFR<45 mL/min/1.73m$^2$, confirmed proteinuria, diabetics with preexisting CKD (GFR<60 mL/min/1.73m$^2$), or whenever the GFR is expected to be <30 mL/min/1.73m$^2$ after intervention.

## Pathology

1. **d. Undergo partial nephrectomy.** Notice the clear cytoplasm due to glycogen characteristic of clear cell carcinoma of the kidney. Because of the patient's young age, a nephron-sparing approach is preferred. Because the tumor is located in the lower pole, is

peripheral, and is exophytic, it is ideal for this approach. These tumors are not particularly sensitive to radiation therapy, cryotherapy is not as reliable for local control, and targeted systemic therapy is not appropriate for a surgically curable lesion.

2. **e. Be followed closely because the development of metastatic disease is likely.** Collecting duct cancers have a high likelihood of being metastatic at diagnosis. None of the other therapies listed are helpful in this disease setting. Cis-platinum chemotherapy is a possibility, but in the absence of known metastasis, its role is not well defined.

## Imaging

1. **b. Radical nephrectomy.** The images demonstrate a large mass in the lower pole of the right kidney, with tumor extension into the right renal vein, extending to the junction of the right renal vein with the inferior vena cava. The renal vein is enlarged, and tumor vessels are seen within the thrombus in the renal vein. These findings make radical nephrectomy the best option.
2. **b. Bosniak IV—cystic RCC.** Notice the enhancing nodules within the cyst wall with foci of dystrophic calcification making cystic RCC the most likely diagnosis.

## Chapter review

1. A solid mass on CT scan that enhances more than 15 Hounsfield units is suggestive of RCC.
2. Twenty percent of small solid enhancing masses less than 4 cm on CT are benign, 40% of small solid enhancing masses less than 2 cm on CT are benign.
3. Hyperdense cysts (Bosniak II) contain old blood and are benign.
4. Clear cell RCCs originate from the proximal tubule; oncocytomas and chromophobe RCCs originate from the distal tubule.
5. There is ample evidence for impaired immune surveillance in RCC.

6. Bilateral involvement in RCC either synchronously or metachronously occurs in 2% to 4% of patients.
7. RCC pathologically is classified as clear cell: 70% to 80%; papillary: 10% to 15%; chromophobe: 3% to 5%; collecting duct: less than 1%; and medullary: rare.
8. Alterations in chromosome 3 are common in clear cell renal carcinoma.
9. Papillary renal cell cancer has a tendency to multifocality.
10. Paraneoplastic syndromes include hypercalcemia, hypertension, polycythemia, and hepatic dysfunction (Stauffer syndrome).
11. Stauffer syndrome is a paraneoplastic syndrome associated with RCC that results in elevated liver function tests. If hepatic function does not normalize after nephrectomy, persistent hepatic dysfunction is indicative of persistent disease.
12. Enlarged perirenal lymph nodes noted on CT may be inflammatory, particularly if they are less than 2 cm in diameter. Lymph nodes larger than 2 cm generally contain metastases.
13. Extended lymphadenectomy for renal cell carcinoma has not been shown to be beneficial for the majority of patients over conventional lymphadenectomy, which includes the renal hilar and adjacent paracaval or para-aortic lymph nodes.
14. The World Health Organization and International Society of Urologic Pathology (ISUP) grading system from 1 to 4 is used to grade the renal tumor and is based on nuclear characteristics.
15. There is an increased incidence of renal cell carcinoma in patients with acquired renal cystic disease.
16. RCC involving the vena cava that infiltrates the wall of the vena cava has an extremely poor prognosis.
17. A patient with a tumor thrombus involving the vena cava associated with metastatic regional nodal disease has a very poor prognosis.
18. A patient with a tumor thrombus involving the vena cava in which the nodes are negative and there is no invasion of the vein wall (except the ostia) has a relatively good prognosis.
19. Ipsilateral adrenalectomy as part of a radical nephrectomy is not necessary unless there is CT evidence of adrenal involvement, contiguous spread of the tumor to the adrenal, or large upper pole

renal masses that are adjacent to with concern for involvement of the adrenal gland.

20. The risk for developing recurrent malignant disease is greatest in the first 3 years after surgery.
21. In a partial nephrectomy, the amount of renal parenchyma taken with the tumor does not affect oncologic outcomes provided the margin itself is negative.
22. RCCs less than 3.5 cm in general grow less than 0.5 cm per year; some may grow up to 1 cm per year.
23. Incomplete excision of a large primary tumor or debulking is rarely indicated as a sole treatment for patients with metastatic kidney cancer.
24. Sarcomas typically have a pseudocapsule that cannot be relied on for a plane of dissection because microscopic tumor will be left behind.
25. Metastatic tumors to the kidney are common, appearing in 12% of patients who die of other cancers; the most common primary lesions are those of the lung, breast, or gastrointestinal tract, melanoma, or hematologic.
26. Major poor prognostic indicators include tumors that extend beyond Gerota fascia to involve contiguous organs (pT4), have lymph node involvement (N1), or are metastatic (M1).
27. Patients with von Hippel-Lindau disease should have all tumors in a kidney treated when the largest reaches 3 cm in size.
28. Patients with a significant reduction in renal mass are at risk for developing long-term renal functional impairment from hyperfiltration renal injury. These patients should have their urinary protein excretion monitored because proteinuria is the initial manifestation of hyperfiltration injury.
29. Established critical prerequisites for successful cryosurgery include rapid freezing, gradual thawing, and a repetition of the freeze-thaw cycle.
30. Approximately 20% of angiomyolipomas are found in patients with the tuberous sclerosis (TS) syndrome, an autosomal dominant disorder characterized by mental retardation, epilepsy, and adenoma sebaceum, a distinctive skin lesion. TS, similar to VHL, is transmitted in an autosomal dominant manner.

31. Cystic nephromas are benign renal neoplasms that occur most commonly in middle-aged women.
32. The VHL syndrome's major manifestations can include the development of renal cell carcinoma, pheochromocytoma, retinal angiomas, and hemangioblastomas of the brain stem, cerebellum, or spinal cord.
33. Inactivation or mutation of the VHL gene leads to dysregulated expression of hypoxia inducible factor, an intracellular protein that plays an important role in regulating cellular responses to hypoxia, starvation, and other stresses. This in turn leads to a several-fold upregulation of the expression of vascular endothelial growth factor (VEGF).
34. Hereditary papillary renal cell carcinoma has an autosomal dominant mode of transmission.
35. Chromosome 3 alterations and VHL mutations are common in clear renal cell carcinoma, and mutation or inactivation of this gene (VHL) has been found in over 75% of sporadic cases.
36. Renal medullary carcinoma is a rare histologic subtype of RCC that occurs almost exclusively in association with sickle cell trait.
37. Most collecting duct carcinomas are centrally located and exhibit an infiltrative growth pattern and aggressive clinical course.
38. Direct ipsilateral adrenal involvement is now grouped with other RCCs that extend beyond Gerota fascia as pathologic stage T4 (pT4).

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[a] Sources referenced can be found in *Campbell-Walsh Urology, 11th Edition,* on the Expert Consult website.